生态研究前沿速览 _2026-04-18

生态研究前沿速览

2026-04-18

共收录 60 篇最新且重要研究

【环境与气候】9 篇 | 【植物与农业】12 篇 | 【生态学核心】12 篇 | 【保护与生物多样性】6 篇 | 【土壤与生物地球化学】3 篇 | 【综合顶刊】4 篇 | 【海洋与水生】10 篇 | 【遥感与模型】4 篇

目录

16. Climate-driven erosion of richness and evolutionary diversity and cons（气候变化正在侵蚀蝾螈热点地区的丰富度、进化多样性与保护成效）

17. Bio-based evaporators for circular agriculture（面向循环农业的生物基蒸发器）

18. Warming Reduces the Positive Effect of Nitrogen Addition on Soil Organ（变暖削弱了氮添加对草地土壤有机碳的正效应）

19. Machine-learning emergent constraints on surface albedo feedback over（机器学习约束下的北极陆地表面反照率反馈强度低于以往估计）

20. Monitoring Coral Reef Metabolism Under Changing Oceans–Novel Insights（利用海水稳定碳同位素监测变化海洋中的珊瑚礁代谢）

21. Energy-mediated feedbacks of vegetation greening enhance precipitation（植被变绿通过能量反馈提高降水效率并维持全球半干旱区水量产出）

22. Wood density variation in European forest species: drivers and implica（欧洲森林树种木材密度变异及其对多尺度生物量与碳评估的影响）

23. Lake Water Depth and Nutrient Gradients Mediate Aquatic Plant Distribu（水深与营养梯度通过性状网络关联强度调节水生植物分布）

24. The role of biofouling and microbial colonization in shaping macroplas（生物污损与微生物定殖如何塑造淡水中大块塑料的归趋）

25. Implementing a Dynamic Root Distribution Scheme Responsive to Soil Env（将响应土壤环境的动态根系分布方案引入 Noah-MP-Crop 以改）

26. Multi-decadal resilience (2002–2023) of subtropical coral communities（尽管遭受累积性强扰动，澳大利亚东部亚热带珊瑚群落在 2002–202）

27. From species to therapeutic property: distribution differences of thre（在气候变化下，受威胁中国药用植物的分布格局及其药效类别差异正在重组）

28. Environmental Fluctuations Are Insufficient to Explain High Coral Biod（环境波动不足以解释高珊瑚多样性：五种储存效应路径检验）

29. Hydrological Mismatch in Arid Planted Shrublands: Non-Responsiveness t（干旱人工灌丛中的水文错配：对降水变化不敏感与不可持续用水）

30. Environmental Variability Shapes Life-History Trade-Offs Within and Be（环境变异塑造长寿海鸟种群内外的生活史权衡）

31. Losses of ecological and social sustainability among small-scale fishe（肯尼亚近岸小规模渔业中的生态与社会可持续性损失）

32. Advances in Modelling Radiative Transfer, Heat Storage and Turbulent T（CanVeg2 模型改进阔叶林二氧化碳、热量与水分通量模拟）

33. Historical and contemporary genomes of an endangered rodent reveal shi（濒危啮齿动物历史与现代基因组揭示环境关联基因变化）

34. A renewable tango for drought-stricken power grids（海洋与水生研究：A renewable tango for drou）

35. Inorganic Carbon (Calcite and Dolomite) Determination Using the Modifi（土壤与生物地球化学研究：Inorganic Carbon (Calc）

36. The linkage between microbial community dynamics and urbanization age（微生物群落动态与城市化年限之间的联系）

37. A dual sustainability crisis for Kenya’s small-scale fisheries（肯尼亚小规模渔业面临双重可持续性危机）

38. Grazing Stocking Rate and Rainfall Jointly Regulate Soil Methane Uptak（放牧载畜率与降雨共同调节温带草甸草原土壤甲烷吸收）

39. Climate as the Main Driver of Large-Scale Genetic Patterns and Connect（气候是扩张中金豺大尺度遗传格局和连通性的主要驱动因子）

40. Decoupled Climatic Drivers of Tree and Ground-Layer Carbon Uptake in M（全球山地生态系统乔木层与地被层碳吸收的气候驱动相互脱耦）

41. Sexual Selection Associated With an Aggressive Male Phenotype Reduces （生态学核心研究：Sexual Selection Associate）

42. Urban water network upgrades improve quality and access to drinking wa（海洋与水生研究：Urban water network upgrad）

43. The Direct and Legacy Impacts of Droughts on Vegetation Growth in Chin（干旱对中国植被生长的直接与滞后影响）

44. Early Postfire Regeneration Determines Long-Term Successional Trajecto（植物与农业研究：Early Postfire Regeneratio）

45. Multiple Mechanisms Required to Predict Grass Community Composition（预测草地群落组成需要多个机制共同支撑）

46. Performative planning creates a values mismatch between wildfire plans（表演式规划造成野火计划与社区需求之间的价值错配）

47. Conserved DNA architect couples chloroplast development to cell cycle （植物与农业研究：Conserved DNA architect co）

48. Global food security rests on the Strait of Hormuz（植物与农业研究：Global food security rests）

49. Stress Dominance and Niche Conservatism in Epiphytic and Lithophytic B（植物与农业研究：Stress Dominance and Niche）

50. Group Size Dependent Selection for Cooperation Versus Freeloading in C（生态学核心研究：Group Size Dependent Selec）

51. Greening the scars（植物与农业研究：Greening the scars）

52. Climatically Driven Stability in the Alpine/Moorland Boundary on Three（植物与农业研究：Climatically Driven Stabil）

53. Benthic diatoms and phytoplankton diversity in urban streams and ponds（城市溪流与池塘中底栖硅藻和浮游植物多样性的保护启示）

54. Thermal stress restructures volatile gas emissions from the model cnid（热胁迫重构模式刺胞动物 Aiptasia 的挥发性气体排放）

55. Drone imaging can accurately assess coral cover, bleaching, and growth（无人机影像可准确评估浅水珊瑚礁覆盖度、白化和生长型）

56. Increasing extremity and accelerating expansion of heatwaves across Eu（欧亚大陆热浪极端性增强且扩张加速）

57. Unveiling shifting frontiers of tropical dry forests depends on consis（保护与生物多样性研究：Unveiling shifting fron）

58. Robots for ecological monitoring: a review（保护与生物多样性研究：Robots for ecological m）

59. Fine-scale aquatic eDNA sampling reveals significant within- and acros（海洋与水生研究：Fine-scale aquatic eDNA sa）

60. Sixty years of artificial lake heating impact Daphnia fitness but not（海洋与水生研究：Sixty years of artificial ）

环境与气候

1. The underappreciated importance of small wetlands in global methane emissions

（全球甲烷排放中小型湿地的重要性被长期低估）

Nature Climate Change | 2026-04-08 | 【Recent】

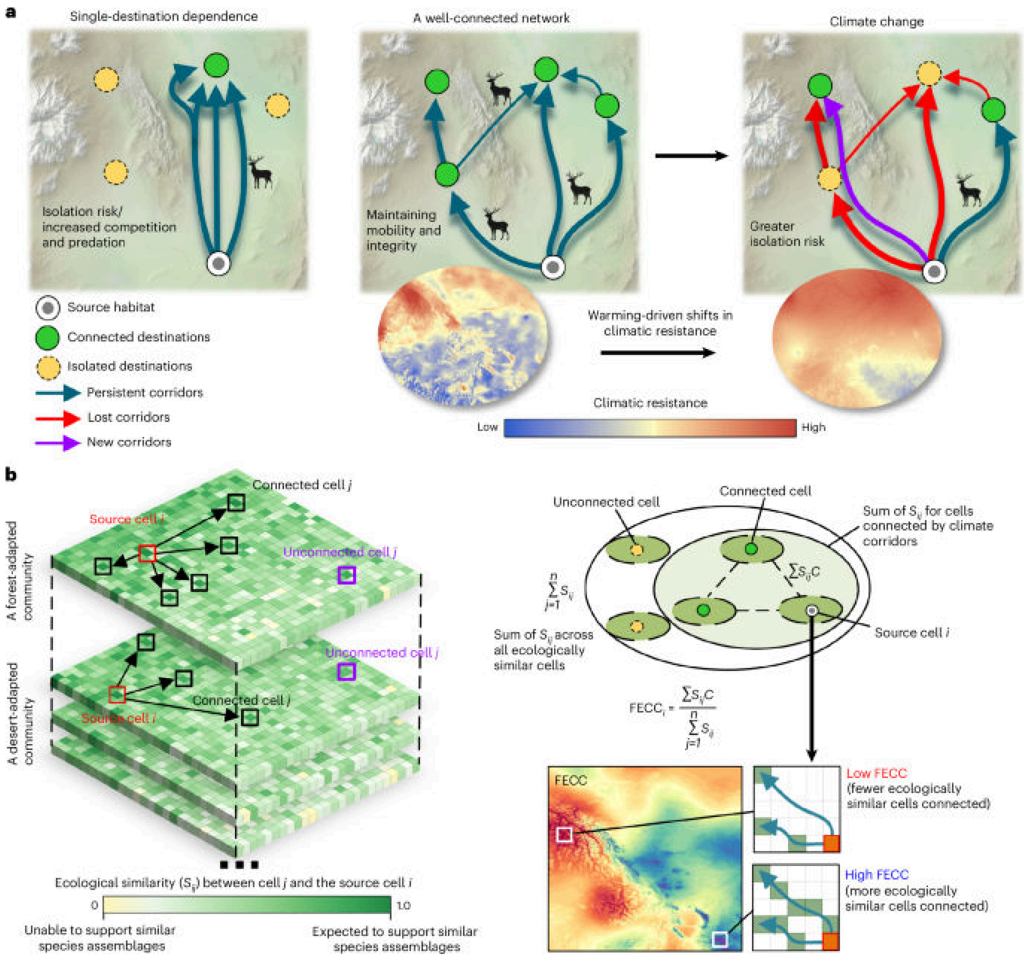


图:Graphical abstract / article figure source: https://media.springernature.com/m685/springer-static/image/art%3A10.1038%2F

Abstract

Nature Climate Change, Published online: 08 April 2026; doi:10.1038/s41558-026-02609-wWetland methane emissions are a major source of uncertainty in global emissions estimates. Here the authors use high-resolution remote sensing data to identify small non-forested wetlands and find that they contribute 24% of wetland methane emissions and that these emissions are increasing.

中文摘要

湿地甲烷排放是全球温室气体估算中的主要不确定来源。研究利用高分辨率遥感数据识别了大量小型非森林湿地，发现这类湿地贡献了约 24% 的湿地甲烷排放，而且其排放仍在上升。

深度解读

这篇文章最关键的提醒是：过去很多全球甲烷核算把面积小、分布碎的湿地低估了。小湿地虽然单体不大，但数量多、排放密集，累积贡献并不"小"。这意味着甲烷减排政策、湿地修复优先区识别以及全球碳预算都需要更高空间分辨率的数据支撑。对生态修复而言，未来不能只盯大型湿地，碎片化湿地同样值得重点管理。

DOI: https://doi.org/10.1038/s41558-026-02609-w

环境与气候

2. Ocean warming weakens the sea–land breeze in coastal megacities

(海洋变暖会削弱沿海特大城市的海陆风降温效应)

Nature Climate Change | 2026-04-17 | 【Latest】

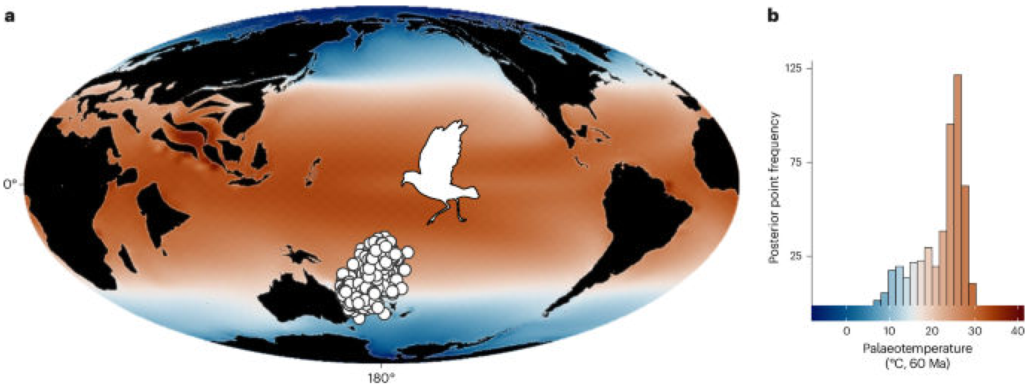


图:Graphical abstract / article figure source: https://media.springernature.com/m685/springer-static/image/art%3A10.1038%2F

Abstract

Nature Climate Change, Published online: 17 April 2026; doi:10.1038/s41558-026-02618-9The sea−land breeze acts to counter urban heat in many coastal cities. Here the authors simulate how this circulation changes with warming ocean water, showing that it decreases in most of them, adding heat stress to urban areas.

中文摘要

研究模拟发现，随着近海海水升温，许多沿海特大城市用于缓解城市热岛的海陆风环流将减弱，从而降低自然通风与降温能力，并进一步加剧城市热暴露风险。

深度解读

很多沿海城市把海风当成'天然空调'，但这篇文章说明这套缓冲机制本身也会被气候变化削弱。问题不只是城市更热，而是原本能够带走热量的近海—陆地环流也在变弱，导致热风险被双重放大。对城市适应来说，这意味着单靠滨海区位并不足以抵消高温，需要把通风廊道、建筑布局和高温应急系统一起纳入规划。对上海、广州、深圳这类超大沿海城市尤其有现实意义。

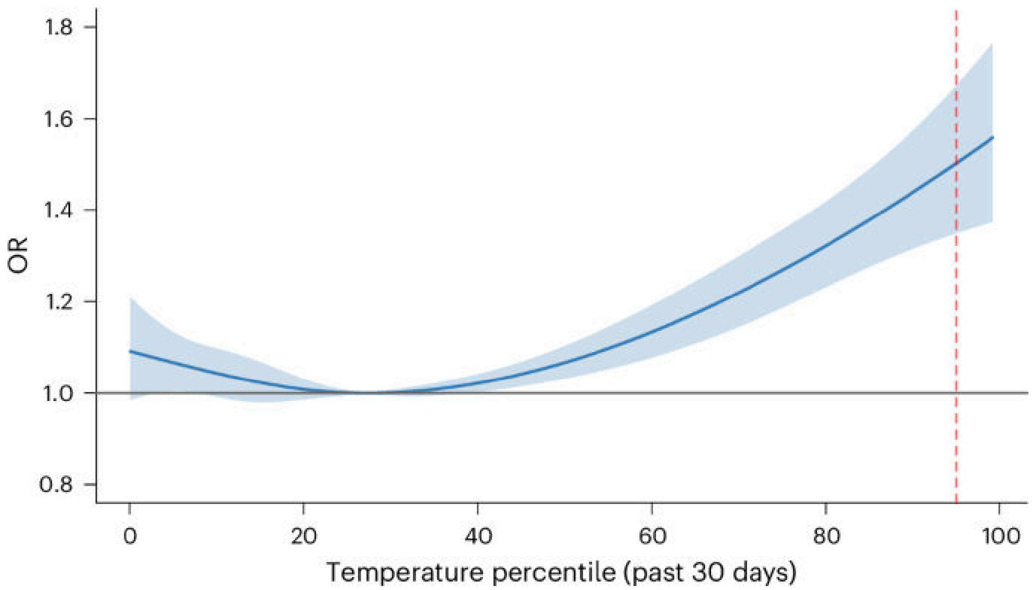
DOI: https://doi.org/10.1038/s41558-026-02618-9

植物与农业

3. Climate change will likely increase the need for heat stress mitigation in apple production

(气候变化可能显著增加苹果生产中热胁迫缓解的需求)

Nature Food | 2026-04-09 | 【Recent】



图：Graphical abstract / article figure source: https://media.springernature.com/m685/springer-static/image/art%3A10.1038%2F

Abstract

Nature Food, Published online: 09 April 2026; doi:10.1038/s43016-026-01343-yA climate-yield model of French apple production quantifies the potential for future climate change adaptation in this sector. While cooler regions can still reap benefits from a warming climate in the short term, yield losses will become increasingly frequent by 2050 due to the lack of adaptive capacity to heat stress.

中文摘要

基于法国苹果生产的气候—产量模型，研究量化了该产业未来适应气候变化的潜力。短期内较冷地区仍可能从适度变暖中受益，但到 2050 年前后，由于热胁迫适应能力不足，减产事件将显著增多。

深度解读

这不是一篇简单说‘变暖有害’的论文，它指出苹果产业最初可能享受一点变暖红利，但很快会被热胁迫反噬。换句话说，果树农业面对的不是线性变化，而是先利后弊的转折过程。文章对果园管理的启示很直接：品种更新、遮阴降温、灌溉优化和区域布局调整都需要提前做，而不是等到高温成为常态再补救。对温带果业尤其具有预警意义。

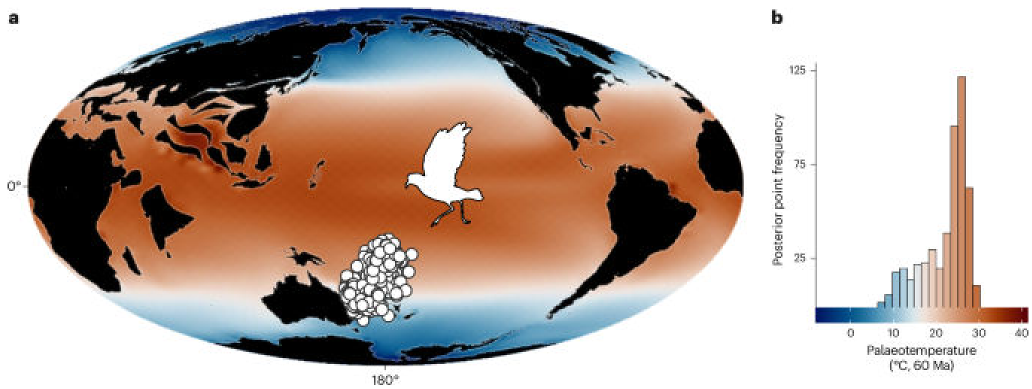
DOI: https://doi.org/10.1038/s43016-026-01343-y

环境与气候

4. From least-cost to SDG-optimal sectoral allocation of Paris Agreement-compatible mitigation efforts

（从最低成本走向 SDG 最优的巴黎协定减排部门分配）

Nature Climate Change | 2026-04-07 | 【Recent】



图：Graphical abstract / article figure source: https://media.springernature.com/m685/springer-static/image/art%3A10.1038%2F

Abstract

Nature Climate Change, Published online: 07 April 2026; doi:10.1038/s41558-026-02602-3Meeting global temperature targets requires deep mitigation across sectors. Moving away from cost optimality when allocating mitigation by sector, the authors link integrated assessment models and portfolio analysis to identify and balance trade-offs between Sustainable Development Goal indicators.

中文摘要

实现全球温控目标需要各部门深度减排。研究将综合评估模型与投资组合分析结合起来，表明如果不再只追求最低经济成本，而是同时权衡可持续发展目标（SDGs），就能得到更平衡的部门减排分配方案。

深度解读

传统减排路径常把“最低成本”当成默认最优，但现实政策从来不只有一个目标。本文的重要价值在于把减排与粮食、安全、健康、公平等 SDG 指标放在同一张表里重新算账。结果说明，单纯追求最便宜的减排路径，可能把生态和社会代价转移到别的部门或群体。对中国和其他发展中经济体来说，这种多目标分配框架比单一成本最优更接近真实政策场景。

DOI: https://doi.org/10.1038/s41558-026-02602-3

植物与农业

5. Designing where to crop and where to graze: a spatial approach toward sustainable farming

（用空间优化重新划定种植区与放牧区以推动可持续农业）

Agronomy for Sustainable Development | 2026-04-14 | 【Latest】

图：Graphical abstract / article figure source: https://static-content.springer.com/image/art%3A10.1007%2Fs13593-026-01096-9

Abstract

The world's growing population raises concerns about future food security. At the same time, environmental challenges such as climate change, biodiversity loss, and the depletion of natural resources must be addressed, calling for a transformation of agriculture. This need is exacerbated by the limited availability of land suitable for arable farming. In this study, we examined the potential of agricultural land for arable farming and grassland use in Switzerland, under the premise that agricultural land use should align with the biophysical and environmental capacity of each location. In an iterative co-design process with scientists and public authorities, we elaborated three scenarios for agricultural transformation, which progressively incorporated (i) biophysical constraints (soil, cl

中文摘要

研究以瑞士为例，依据不同地块的生物物理条件与环境承载力，评估哪些农业用地更适合耕作、哪些更适合草地与放牧。结果表明，将土地利用与地点适宜性更精确匹配，有望在粮食安全、资源效率和生态环境之间取得更优平衡。

深度解读

这篇文章的价值在于，它不再把农业问题理解成'多生产一点就行'，而是追问：不同地方究竟最适合做什么。把耕作放在真正适宜耕作的地块，把放牧留给更适合草地利用的区域，本质上是在做农业空间重构。这样的思路既有助于减少低效投入，也能减轻土壤退化、养分流失和生境压力。对山地农业、草牧交错带和国土空间优化来说，这是一种很值得推广的方法框架。

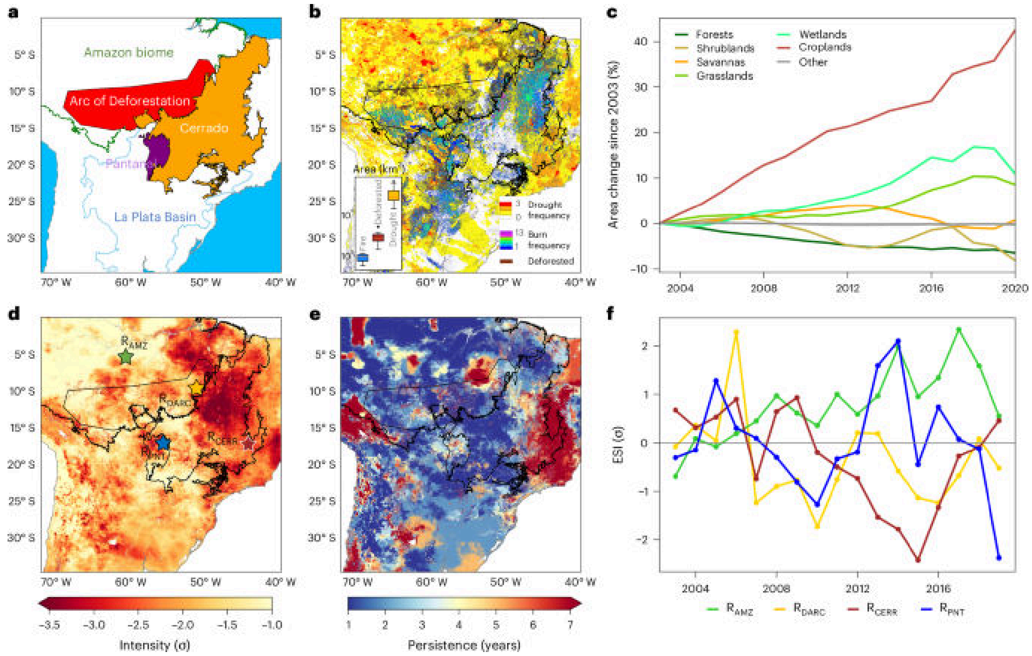
DOI: https://link.springer.com/article/10.1007/s13593-026-01096-9

环境与气候

6. Aligning climate change mitigation strategies with policy objectives beyond cost savings

（超越成本最优：让气候减缓路径与多重政策目标对齐）

Nature Climate Change | 2026-04-15 | 【Latest】



图：Graphical abstract / article figure source: https://media.springernature.com/m685/springer-static/image/art%3A10.1038%2F

Abstract

Nature Climate Change, Published online: 15 April 2026; doi:10.1038/s41558-026-02617-wOptimal climate change mitigation pathways have historically focused on achieving emissions reductions while ensuring cost efficiency. However, the broader impacts of climate action are also important for policymakers and stakeholders. We developed a method that enables mitigation pathways to be defined based on their impact on multiple United Nations Sustainable Development Goals (SDGs).

中文摘要

研究提出一种新的路径设计方法，不再只以减排成本最低为目标，而是把多个联合国可持续发展目标（SDGs）的影响纳入评估。结果显示，兼顾发展、公平与环境协同效益，可以得到与传统成本最优路径显著不同的减排策略组合。

深度解读

传统减排路径常把'最低成本'当成默认最优，但现实政策从来不只有一个目标。本文的重要价值在于把减排与粮食、安全、健康、公平等 SDG 指标放在同一张表里重新算账。结果说明，单纯追求最便宜的减排路径，可能把生态和社会代价转移到别的部门或群体。对中国和其他发展中经济体来说，这种多目标分配框架比单一成本最优更接近真实政策场景。

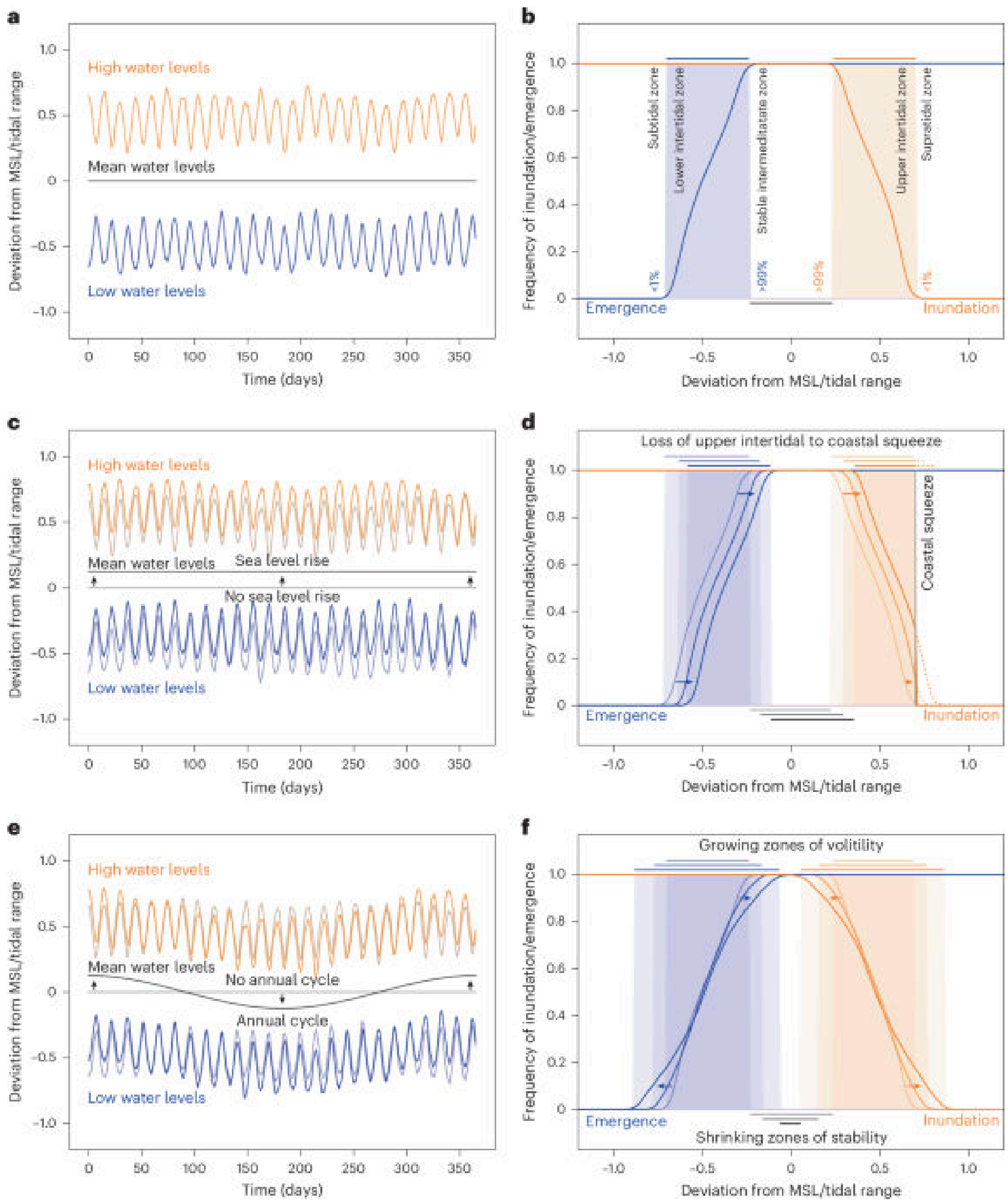
DOI: https://doi.org/10.1038/s41558-026-02617-w

生态学核心

7. Cross-ecosystem linkages between freshwater insects and riparian birds across the USA

(美国尺度上淡水昆虫与河岸鸟类之间的跨生态系统联系)

Nature Ecology & Evolution | 2026-04-07 | 【Recent】



图：Graphical abstract / article figure source: https://media.springernature.com/m685/springer-static/image/art%3A10.1038%2F

Abstract

Nature Ecology & Evolution, Published online: 07 April 2026; doi:10.1038/s41559-026-03041-1Aquatic insects such as mayflies, stoneflies and caddisflies are widely used for freshwater biomonitoring and could provide resource pulses to terrestrial insectivores. Here the authors show a positive relationship between the species richness of these insects and the prevalence of riparian birds across the conterminous USA.

中文摘要

蜉蝣、石蝇和石蛾等水生昆虫常被用于淡水生物监测，也可能为陆地食虫动物提供关键资源脉冲。研究表明，在美国本土范围内，这些淡水昆虫的物种丰富度越高，河岸鸟类出现的概率也越高。

深度解读

这项研究把“水域健康”和“岸上鸟类”直接连了起来。它说明河流和陆地并不是两个孤立系统，水生昆虫羽化后把能量和营养带上岸，支撑了河岸鸟类群落。对流域治理来说，这意味着修复水环境不仅是在保护鱼和无脊椎动物，也是在修复更大的陆—水交错生态网络。很多看似是鸟类保护问题，源头可能其实在河流。

DOI: https://doi.org/10.1038/s41559-026-03041-1

环境与气候

8. More eddying of subtropical western boundary currents boosts stratification and cools shelf seas

（副热带西边界流涡旋增强会加剧层化并冷却陆架海）

Nature Climate Change | 2026-04-15 | 【Latest】

图：Graphical abstract / article figure source: https://media.springernature.com/m685/springer-static/image/art%3A10.1038%2F

Abstract

Nature Climate Change, Published online: 15 April 2026; doi:10.1038/s41558-026-02599-9This work shows that increased eddies accelerate surface warming in the Agulhas Current while also boosting hidden upwelling that cools the current, and adjacent shelf seas, at depth. Similar trends are expected for all subtropical western boundary currents, even if volume transports remain steady.

中文摘要

研究显示，副热带西边界流中更频繁或更强的中尺度涡旋，一方面会加速表层变暖，另一方面也会增强隐蔽上升流，使主流及邻近陆架海在深层出现冷却并加强水体层化。这种现象预计不只发生在阿古拉斯洋流，也可能是类似洋流系统的普遍趋势。

深度解读

这项工作很有意思，因为它揭示了海洋变暖并不总是表现为'整层一起变热'。涡旋增强会把表层增温、深层冷却和层化加强同时推高，导致近岸陆架海的生态环境更复杂。对渔业、营养盐输送和沿海低氧风险来说，这种垂向结构变化往往比平均温度本身更关键。它提醒海洋管理不能只盯海表温度，必须把中尺度过程和近岸生态响应一起看。

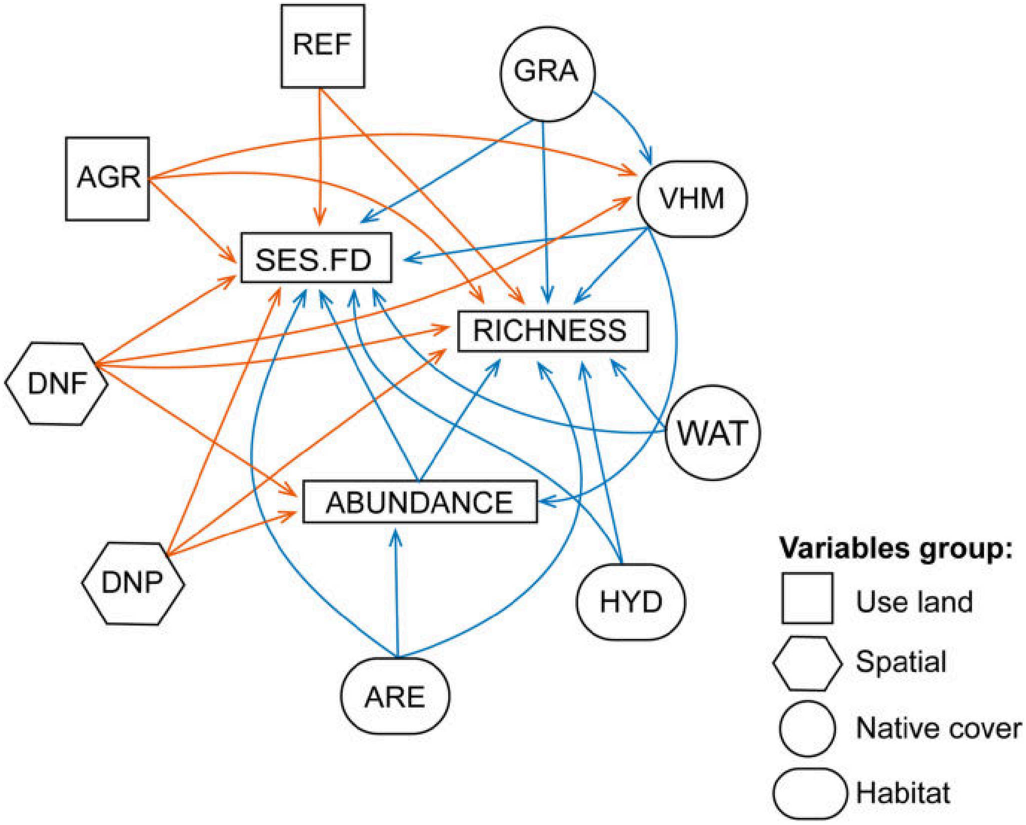
DOI: https://doi.org/10.1038/s41558-026-02599-9

保护与生物多样性

9. Distribution of the baraúna tree is threatened by climate change, land-cover change, and biological invasions in the Brazilian Caatinga

（气候变化、土地覆被变化与生物入侵共同威胁巴西旱林珍稀树种 baraúna）

Biodiversity and Conservation | 2026-04-16 | 【Latest】



图：Graphical abstract / article figure source: https://static-content.springer.com/image/art%3A10.1007%2Fs10531-026-03340-w

Abstract

Multiple drivers shape biodiversity loss worldwide. Land-use change and biological invasions are major direct pressures on species, while climate change increasingly alters species distributions and community dynamics, often intensifying other stressors. However, these factors are still insufficiently integrated into Ecological Niche Modeling (ENM) studies. We quantified the potentially suitable area for Schinopsis brasiliensis Engl., a tree species of high ecological importance in the Caatinga domain, considering the combined effects of climate change, land-cover change (LCC), and invasion by the exotic species Neltuma juliflora (Sw.) Raf. (hereafter, Neltuma spp.), known to reduce native plant abundance. We modeled the potential distributions of S. brasiliensis and Neltuma spp. under cur

中文摘要

研究将气候变化、土地覆被变化及生物入侵同时纳入生态位模型，评估巴西卡廷加地区关键树种 Schinopsis brasiliensis 的潜在适生区变化。结果显示，多重压力叠加将明显压缩其未来分布范围，并提高保护与恢复难度。

深度解读

这篇文章的重要之处在于，它没有把威胁因素拆开单独讨论，而是把气候、土地利用和入侵种放在同一个风险框架里。现实中的物种衰退往往就是这样发生的：不是单一因子致命，而是多重压力联手挤压。对于旱区树种保护而言，这意味着保护工作不能只做保育地划定，还要同步考虑土地管理和入侵防控。文章也说明，未来的分布模型如果仍然只看气候，往往会低估真正的灭失风险。

DOI: https://link.springer.com/article/10.1007/s10531-026-03340-w

---

土壤与生物地球化学

10. Understanding the balance between methane production and oxidation from wetlands: insights from a reduced process-based model

(理解湿地甲烷产生与氧化平衡：来自简化过程模型的洞见)

Biogeosciences | 2026-04-09 | 【Recent】

图：Graphical abstract / article figure source:

Abstract

Understanding the balance between methane production and oxidation from wetlands: insights from a reduced process-based model Gordon R. McNicol, Anita T. Layton, and Nandita B. Basu Biogeosciences, 23, 2309–2333, https://doi.org/10.5194/bg-23-2309-2026, 2026 We have developed a simple mathematical model to investigate how water levels, soil temperature, and plant carbon inputs control methane release from wetlands through production and consumption in the soil. The model captures seasonal and inter-annual variations in emissions and identifies conditions, particularly water depth, that maximise emissions, depending on wetland characteristics. This work supports predicting greenhouse gas fluxes and guiding wetland management and restoration.

中文摘要

研究建立了一个简化的过程模型，用于分析水位、土壤温度和植物碳输入如何共同控制湿地甲烷的产生、氧化与释放。模型能够较好再现季节和年际变化，并指出不同湿地类型下最关键的控制因素往往是水位深度。

深度解读

湿地甲烷排放之所以难管，很大程度上是因为“产生”和“氧化”两个过程同时在变。本文的价值不在于做得更复杂，而是用一个足够简洁但机制清楚的模型，把关键控制因子抓了出来。它尤其提示管理者：水位调控往往比想象中更有杠杆作用。对于湿地修复和温室气体协同管理，这类可解释模型比黑箱预测更实用。

DOI: https://doi.org/10.5194/bg-23-2309-2026

植物与农业

11. Climate change and crop-feeding herbivores

(气候变化与取食作物的植食性动物风险)

Nature Food | 2026-04-17 | 【Latest】

图：Graphical abstract / article figure source: https://media.springernature.com/m685/springer-static/image/art%3A10.1038%2F

Abstract

Nature Food, Published online: 17 April 2026; doi:10.1038/s43016-026-01346-9Climate change and crop-feeding herbivores

中文摘要

研究围绕“气候变化与取食作物的植食性动物风险”展开。英文摘要要点为：Climate change and crop-feeding herbivores 该条目发表于 Nature Food，报告中保留 DOI 与原文链接，便于后续核对全文细节。

深度解读

该研究连接植物功能、农业生产和环境胁迫，对气候适应型农业与生态系统功能评估有启发。其重要性在于把具体观测或模型结果转化为可检验的机制判断。对 2026-04-18 这期速览而言，它补充了当周关于植物与农业方向的前沿证据。后续使用时应结合全文方法、空间尺度和不确定性，避免把单一区域或单一模型结论过度外推。

DOI: https://doi.org/10.1038/s43016-026-01346-9

环境与气候

12. Humanitarian blind spots in Western climate change policy and discourse

（环境与气候研究：Humanitarian blind spots in Western climate change policy and discourse）

Nature Climate Change | 2026-04-16 | 【Latest】

图：Graphical abstract / article figure source:

Abstract

Nature Climate Change, Published online: 16 April 2026; doi:10.1038/s41558-026-02613-0Humanitarian blind spots in Western climate change policy and discourse

中文摘要

研究围绕“环境与气候研究：Humanitarian blind spots in Western climate change policy and discourse”展开。英文摘要要点为：Humanitarian blind spots in Western climate change policy and discourse 该条目发表于 Nature Climate Change，报告中保留 DOI 与原文链接，便于后续核对全文细节。

深度解读

该研究与气候变化、极端事件或环境治理直接相关，能帮助判断生态系统和人类社会面对的复合风险。其重要性在于把具体观测或模型结果转化为可检验的机制判断。对 2026-04-18 这期速览而言，它补充了当周关于环境与气候方向的前沿证据。后续使用时应结合全文方法、空间尺度和不确定性，避免把单一区域或单一模型结论过度外推。

DOI: https://doi.org/10.1038/s41558-026-02613-0

环境与气候

13. 15 years of change

（环境与气候研究：15 years of change）

Nature Climate Change | 2026-04-10 | 【Recent】

图：Graphical abstract / article figure source: https://www.nature.com/immersive/s41558-026-02610-3/assets/tVyz9wHF19/smca

Abstract

Nature Climate Change, Published online: 10 April 2026; doi:10.1038/s41558-026-02610-3Since Nature Climate Change was launched, not only has the journal itself changed but so have the subjects of the studies we publish on the Earth system and how societies interact with it. In this Infographic, we highlight a few examples of how the world differed when we started in 2011 compared with today.

中文摘要

研究围绕"环境与气候研究：15 years of change"展开。英文摘要要点为：Since Nature Climate Change was launched, not only has the journal itself changed but so have the subjects of the studies we publish on the Earth system and how societies interact with it. In this Infographic, we highlight a few examples of how the world differed when we started in 2011 compared with today. 该条目发表于 Nature Climate Change，报告中保留 DOI 与原文链接，便于后续核对全文细节。

深度解读

该研究与气候变化、极端事件或环境治理直接相关，能帮助判断生态系统和人类社会面对的复合风险。其重要性在于把具体观测或模型结果转化为可检验的机制判断。对 2026-04-18 这期速览而言，它补充了当周关于环境与气候方向的前沿证据。后续使用时应结合全文方法、空间尺度和不确定性，避免把单一区域或单一模型结论过度外推。

DOI: https://doi.org/10.1038/s41558-026-02610-3

植物与农业

14. Higher optimal temperature for vegetation transpiration than for photosynthesis

（植被蒸腾的最适温度高于光合作用的最适温度）

Nature Plants | 2026-04-15 | 【Latest】

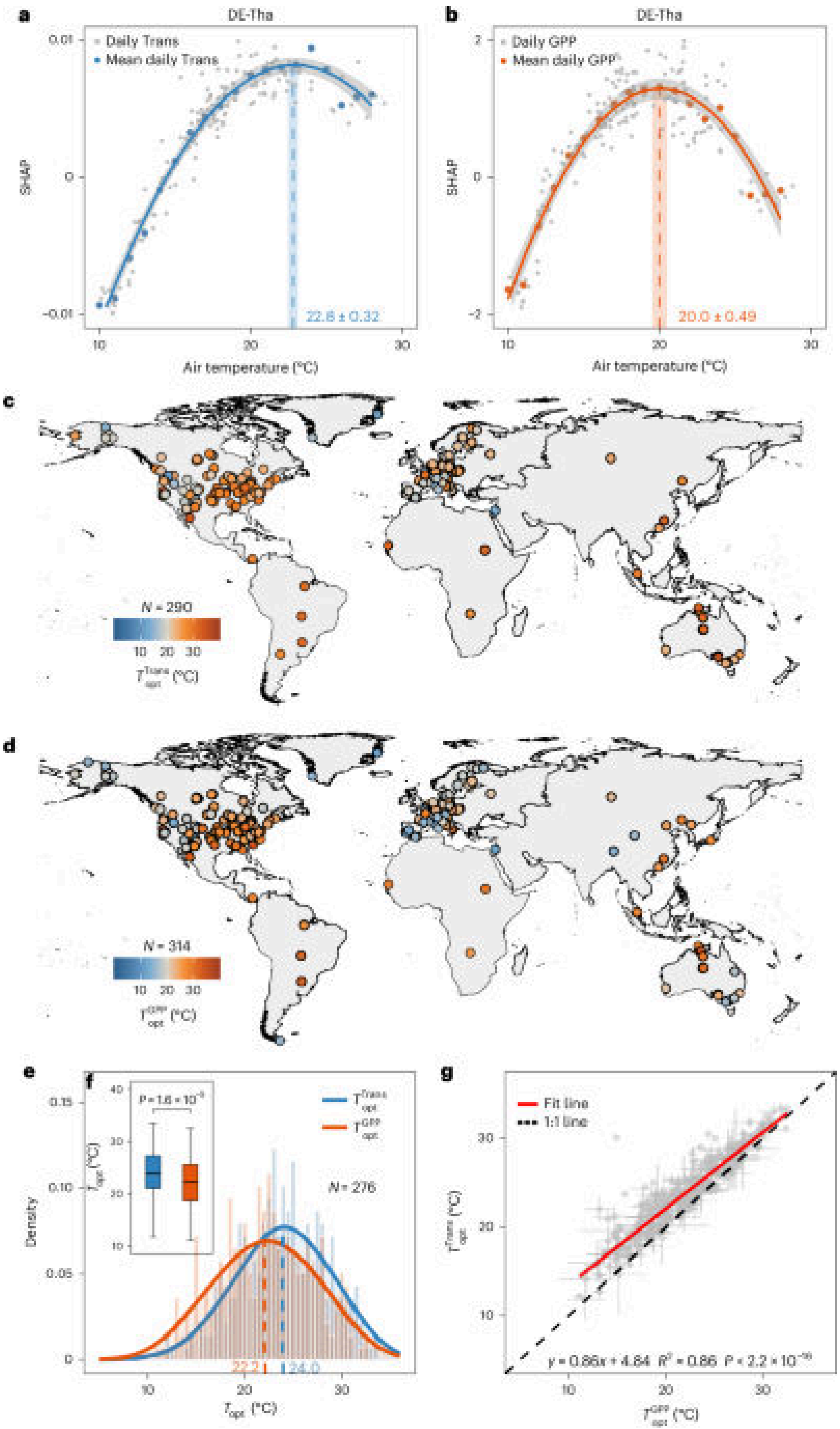


图 : Graphical abstract / article figure source: https://media.springernature.com/m685/springer-static/image/art%3A10.1038%2F

Abstract

Nature Plants, Published online: 15 April 2026; doi:10.1038/s41477-026-02263-2Global observations show that ecosystems sustain transpiration at temperatures above those that maximize photosynthesis. This thermal gap reveals that carbon uptake declines before water loss under climate warming.

中文摘要

全球观测表明，生态系统维持蒸腾作用的最适温度普遍高于光合作用达到峰值的温度。这意味着在持续变暖背景下，碳吸收会先于水分散失开始下降，暴露出生态系统'先失碳、后失水'的热响应差异。

深度解读

这篇 Nature Plants 文章抓住了陆地生态系统对变暖响应中的一个关键不对称性：植物在高温下会先失去固碳效率，却还在继续耗水。换句话说，生态系统可能更早从'高效碳汇'变成'低效用水者'。这对干旱风险评估、作物模型和全球碳循环预测都非常重要，因为很多模型默认碳—水响应是同步的。若这种热阈值错位普遍存在，未来高温对生态系统功能的冲击可能比预想更早到来。

DOI: https://doi.org/10.1038/s41477-026-02263-2

综合顶刊

15. Hydraulic stress limits thermal acclimation in trees under chronic drought

（长期干旱下液压胁迫限制树木对高温的适应性驯化）

PNAS | 2026-04-06 | 【Recent】

图：Graphical abstract / article figure source:

Abstract

SignificanceFuture forest resilience depends on whether trees can acclimate to rising heat and drought. Using multiyear manipulations of soil moisture and air temperature, we tested how long-term acclimation shapes leaf cooling and damage inFagus ...

中文摘要

研究利用多年土壤水分和气温操控实验，检验长期干旱条件下树木是否能通过调节叶片冷却和耐热性来适应升温。结果表明，液压胁迫会显著限制这种热适应能力，从而削弱未来森林在高温干旱环境中的韧性。

深度解读

森林能否在未来继续扛住热浪，关键不只是气温升高多少，还取决于树木有没有足够水力系统支撑散热和维持功能。本文指出，长期干旱会卡住树木的热适应过程，使它们即便经历长期暴露，也难

以真正‘练出耐热性’。这对欧洲和全球温带森林都是个坏消息，因为热浪与干旱往往一起发生。对森林经营而言，水分条件和物种配置可能比单纯追求短期生长更重要。

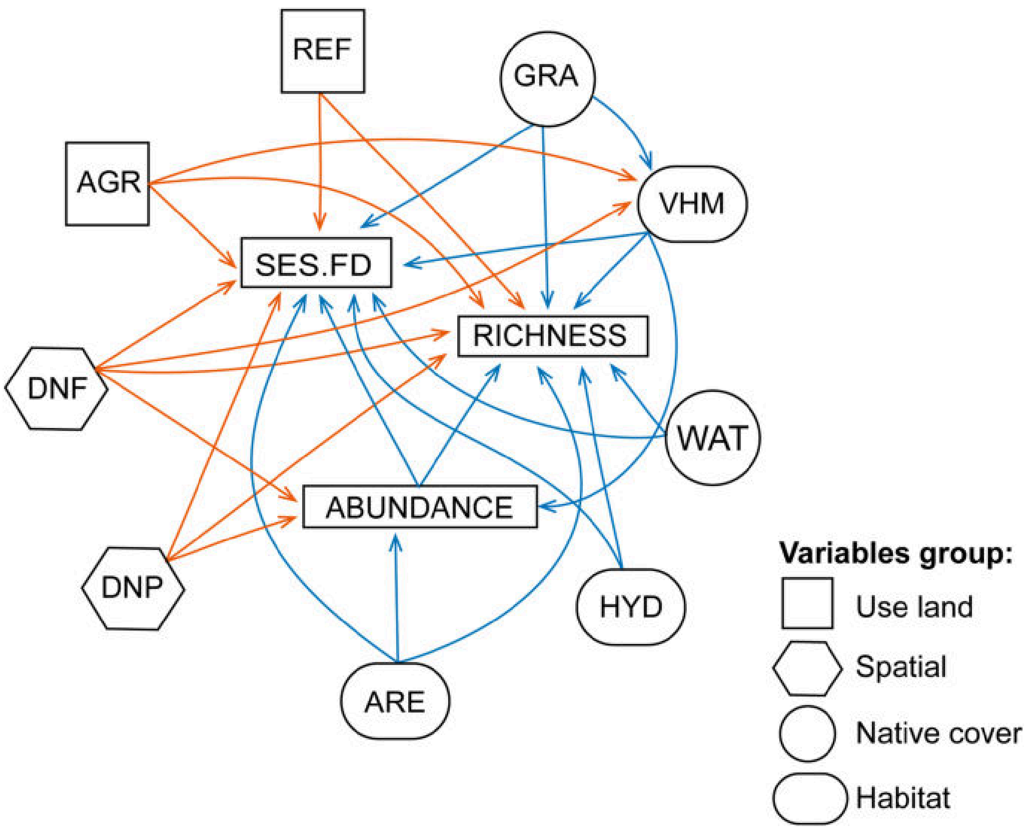
DOI: https://www.pnas.org/doi/abs/10.1073/pnas.2531865123

保护与生物多样性

16. Climate-driven erosion of richness and evolutionary diversity and conservation shortfall in a salamander hotspot

（气候变化正在侵蚀蝾螈热点地区的丰富度、进化多样性与保护成效）

Biodiversity and Conservation | 2026-04-07 | 【Recent】



图：Graphical abstract / article figure source: https://static-content.springer.com/image/art%3A10.1007%2Fs10531-026-03346-

Abstract

Anthropogenic climate change poses a significant threat to biodiversity, particularly for amphibians inhabiting tropical mountain systems. Due to their reliance on humid montane forests and moderate thermal conditions, plethodontid salamanders in Nuclear Central America, a global hotspot of diversity and endemism, are especially vulnerable to warming and altered precipitation regimes. We applied species distribution models (SDM) to project the current and future distributions of 35 salamander species under moderate (SSP2-4.5) and high-emissions (SSP5-8.5) climate scenarios. We quantified changes in range size, species−area relationships, phylogenetic diversity, endemism, and evolutionary distinctiveness, and evaluated biodiversity patterns across protected areas under different governance

中文摘要

气候变化对热带山地两栖动物构成严重威胁。研究以中美洲一个蝾螈多样性热点为对象，基于 35 个物种的分布模型评估不同排放情景下的分布区缩减、系统发育多样性下降、地方性丧失以及保护地覆盖不足等问题。

深度解读

这篇文章值得重视，因为它不是只看“物种数掉了多少”，而是把进化历史一起算进去了。一个地区如果失去的是进化独特的古老支系，其生态和保护代价会远大于简单的物种计数下降。研究还指出现有保护地体系对未来气候位移的覆盖并不充分，说明静态保护网络可能守不住动态气候风险。对山地两栖类保护，这几乎是一个明确的升级信号。

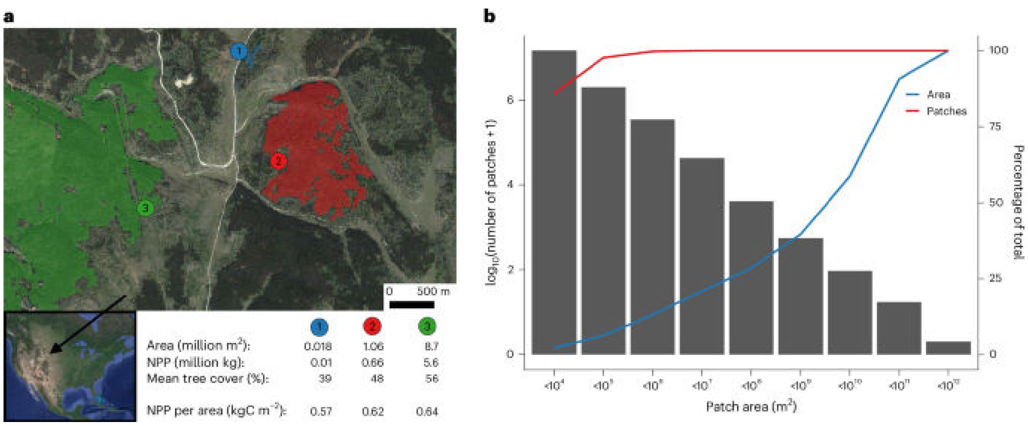
DOI: https://link.springer.com/article/10.1007/s10531-026-03346-4

海洋与水生

17. Bio-based evaporators for circular agriculture

（面向循环农业的生物基蒸发器）

Nature Water | 2026-04-08 | 【Recent】



图：Graphical abstract / article figure source: https://media.springernature.com/m685/springer-static/image/art%3A10.1038%2F

Abstract

Nature Water, Published online: 08 April 2026; doi:10.1038/s44221-026-00627-8Solar-driven interfacial evaporation offers a sustainable path for freshwater production, yet its integration into a truly circular agricultural economy has remained elusive. A study now demonstrates a ‘closed-loop’system using amyloid fibril-based bioevaporators to desalinate water for irrigation, subsequently recycling agricultural waste back into the desalination process.

中文摘要

太阳驱动界面蒸发被视为可持续淡水生产路径，但此前很难真正融入循环农业体系。研究展示了一种基于淀粉样纤维生物蒸发器的闭环系统，可用于灌溉脱盐，并将农业废弃物重新回收进入淡化过程。

深度解读

这项研究好看的地方在于，它不只是做了一个高效率蒸发器，而是把“淡化—灌溉—农业废弃物再利用”串成了闭环。生态农业领域最缺的不是单点技术，而是能减少外部投入、内部循环的系统设计。文章说明，农业废弃物本身可以成为水处理系统的一部分，这对缺水地区尤其有吸引力。若工程放大可行，它会是一类很典型的农业—水资源协同技术。

DOI: https://doi.org/10.1038/s44221-026-00627-8

生态学核心

18. Warming Reduces the Positive Effect of Nitrogen Addition on Soil Organic Carbon in Grasslands

（变暖削弱了氮添加对草地土壤有机碳的正效应）

Ecology Letters | 2026-04-06 | 【Recent】

图： Graphical abstract / article figure source:

Abstract

Nitrogen (N) addition can increase soil organic carbon (SOC) in some grasslands, however, it is unknown whether N addition may enhance SOC in a warmer climate at broad spatial scales. We conduct a field experiment to test how N addition and warming interactively influence SOC in a temperate semiarid grassland. N addition significantly increases SOC by 14%, while simultaneous N addition and warming significantly decreases SOC by 25% relative to N-only addition. This result is further supported by a global meta-analysis, which shows that N addition (1 g N m^{-2} yr^{-1}) increases SOC content by 0.2% relative to ambient conditions in grasslands, but the positive effect of N on SOC significantly declines at a rate of 0.06% per °C of increased mean annual temperature. Our field experiment and meta-analysis suggest that warming may reduce or even eliminate the positive effect of N addition on SOC in grasslands.

中文摘要

氮添加在部分草地中能够提高土壤有机碳，但在更暖气候下这一效应是否仍然成立并不清楚。研究通过半干旱温带草地野外实验和全球荟萃分析发现，单独氮添加可使土壤有机碳增加 14%，而增温与氮添加同时发生时，相对于仅氮添加处理却会使土壤有机碳降低 25%；且氮添加的增碳效应会随年均温升高而持续减弱。

深度解读

这项工作直接修正了一个常见预期：‘加氮就能增加土壤碳储量’并不是放之四海而皆准。随着气候变暖，氮输入带来的增碳收益可能被呼吸增强、水分胁迫或分解加快抵消掉。也就是说，未来很多草地生态系统里，氮沉降未必是碳汇利好，反而可能与变暖叠加后削弱碳存储。对碳汇评估和草地管理来说，这个交互效应不能再忽略。

DOI: https://onlinelibrary.wiley.com/doi/10.1111/ele.70379

综合顶刊

19. Machine-learning emergent constraints on surface albedo feedback over Arctic land regions

（机器学习约束下的北极陆地表面反照率反馈强度低于以往估计）

Nature Communications | 2026-04-10 | 【Recent】

图：Graphical abstract / article figure source:

Abstract

Nature Communications, Published online: 10 April 2026; doi:10.1038/s41467-026-71779-0This study uses machine learning and ground data from the Arctic to constrain land albedo feedback. The projected feedback is reduced by 0.29–0.52 W m^{-2} K^{-1} and uncertainty by 45–55%, suggesting that climate models overestimate this key warming amplifier.

中文摘要

研究结合机器学习与北极地面观测数据，对陆地表面反照率反馈进行了涌现约束。结果显示，这一关键增暖反馈的预测值可降低 0.29–0.52 W m−2 K−1，不确定性也减少了 45%–55%，意味着现有气候模型可能高估了北极陆地反照率反馈。

深度解读

北极变暖为什么这么快，反照率反馈一直是核心机制之一。本文的意义在于，不是简单再跑一次模型，而是用机器学习把模型和观测之间的偏差“掐”得更准。结果显示，部分模型可能把陆地反照率反馈估得过强，这会影响我们对北极放大效应和全球气候敏感度的判断。它不是说北极风险变小了，而是说关键反馈参数需要更精细地重新校准。

DOI: https://doi.org/10.1038/s41467-026-71779-0

遥感与模型

20. Monitoring Coral Reef Metabolism Under Changing Oceans−Novel Insights From Seawater Stable Carbon Isotopes

（利用海水稳定碳同位素监测变化海洋中的珊瑚礁代谢）

Journal of Geophysical Research - Biogeosciences | 2026-04-18 | 【Latest】

图：Graphical abstract / article figure source:

Abstract

Journal of Geophysical Research: Biogeosciences, Volume 131, Issue 4, April 2026.

中文摘要

研究围绕“利用海水稳定碳同位素监测变化海洋中的珊瑚礁代谢”展开。英文摘要要点为：Journal of Geophysical Research: Biogeosciences, Volume 131, Issue 4, April 2026. 该条目发表于 Journal of Geophysical Research - Biogeosciences，报告中保留 DOI 与原文链接，便于后续核对全文细节。

深度解读

该研究围绕群落、种群、性状或生态系统过程，提供了可迁移的生态学机制认识。其重要性在于把具体观测或模型结果转化为可检验的机制判断。对 2026-04-18 这期速览而言，它补充了当周关于遥感与模型方向的前沿证据。后续使用时应结合全文方法、空间尺度和不确定性，避免把单一区域或单一模型结论过度外推。

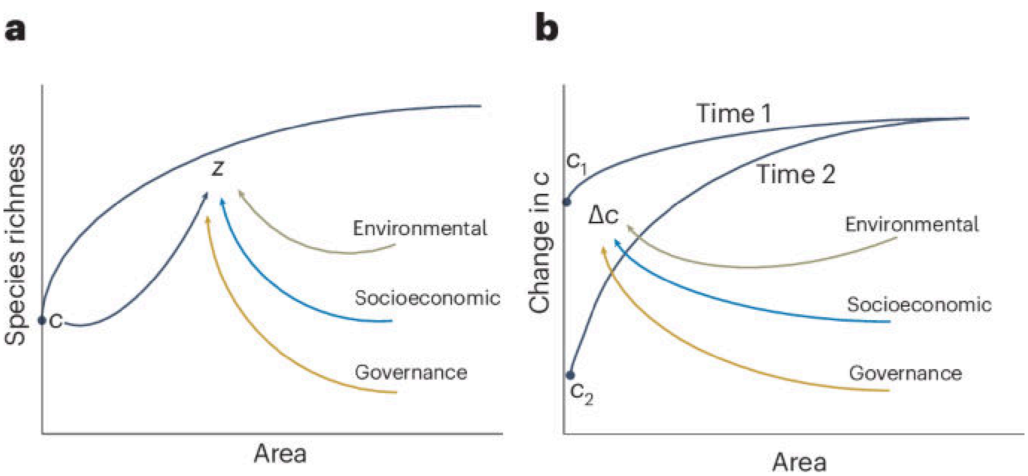
DOI: https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2025JG009416

海洋与水生

21. Energy-mediated feedbacks of vegetation greening enhance precipitation efficiency and sustain water yield in global semi-arid regions

(植被变绿通过能量反馈提高降水效率并维持全球半干旱区水量产出)

Nature Water | 2026-04-17 | 【Latest】



图：Graphical abstract / article figure source: https://media.springernature.com/m685/springer-static/image/art%3A10.1038%2F

Abstract

Nature Water, Published online: 17 April 2026; doi:10.1038/s44221-026-00631-yVegetation greening is traditionally thought to increase evapotranspiration and reduce streamflow, posing a challenge for balancing ecosystem restoration with water supply. This study reveals that greening can enhance both evapotranspiration and streamflow, particularly in semi-arid regions through albedo-driven, energy-mediated increases in the efficiency with which atmospheric moisture is converted into precipitation.

中文摘要

传统观点认为植被恢复会因蒸散增强而减少径流，但该研究发现，在全球半干旱区，植被变绿可通过反照率变化等能量反馈提高大气水汽转化为降水的效率，从而同时提升蒸散和水量产出。

深度解读

这项研究之所以重要，是因为它挑战了‘绿了就更耗水’的单向叙事。文章表明，在半干旱地区，植被恢复并不一定与水资源目标冲突，关键在于能量收支和区域降水反馈如何被重新组织。换句话说，生态修复可能不是简单地从河流‘拿走水’，而是在某些区域反过来提升水循环效率。对退化治理、三北工程和干旱区流域管理来说，这类结果很值得深入跟进。

DOI: https://doi.org/10.1038/s44221-026-00631-y

土壤与生物地球化学

22. Wood density variation in European forest species: drivers and implications for multiscale biomass and carbon assessment in France

(欧洲森林树种木材密度变异及其对多尺度生物量与碳评估的影响)

Biogeosciences | 2026-04-10 | 【Recent】

图：Graphical abstract / article figure source:

Abstract

Wood density variation in European forest species: drivers and implications for multiscale biomass and carbon assessment in France Henri Cuny, Jean-Daniel Bontemps, Nikola Besic, Antoine Colin, Lionel Hertzog, Amaël Le Squin, William Marchand, Cédric Vega, and Jean-Michel Leban Biogeosciences, 23, 2365 – 2388, https://doi.org/10.5194/bg-23-2365-2026, 2026 We analysed wood density variation in mainland France using 110,000 records from 156 tree species. Models calibrated for 44 species revealed species identity as the dominant driver, followed by tree size and temperature. While neglecting wood density variation had little impact on forest biomass estimates nationally, it caused significant bias at finer scales, highlighting the need to account for wood density variation in forest biomass and

中文摘要

研究基于法国本土 156 个树种、11 万条记录分析了木材密度的变化来源。结果表明，树种身份是最主要驱动因子，其次是树木大小和温度；忽视木材密度变异在国家尺度上对生物量估算影响不大，但在更细尺度上会导致显著偏差。

深度解读

森林碳核算常把木材密度当成一个相对固定的参数，但这篇文章提醒我们，尺度一细，误差就会被放大。对于国家级总量估算，平均值可能还凑合；但一旦进入区域经营、样地评估或碳项目核算，木材密度变异就会明显影响结果。它对森林遥感反演、碳储量盘点和经营决策都有现实价值。说白了，‘精细化碳核算’需要更精细的木材性状参数。

DOI: https://doi.org/10.5194/bg-23-2365-2026

生态学核心

23. Lake Water Depth and Nutrient Gradients Mediate Aquatic Plant Distribution via the Strength of Trait Network Associations

（水深与营养梯度通过性状网络关联强度调节水生植物分布）

Global Change Biology | 2026-04-09 | 【Recent】

图：Graphical abstract / article figure source:

Abstract

Aquatic macrophyte responses and adaptation to water depth and nutrient gradients are of critical importance for understanding the functional ecology of lakes and their restoration when degraded. However, the role of plant trait networks (PTNs), a powerful tool for decoding trait−environment−function relationships, in mediating these adaptations remains unclear, particularly across species and community scales. We conducted cross-scale PTN analyses at species and community levels for aquatic macrophytes, integrating assessments of phylogenetic signals and hierarchical environmental filtering. At species level, aquatic macrophytes exhibited two complementary PTN-based adaptive strategies: trait compartmentalization to enhance stress tolerance and trait integration to optimize resource utilization. Notably, local environmental conditions rather than phylogenetic signals dominated the pattern of PTN metrics. At community level, environmental filtering operated hierarchically, acting sequentially by water depth, nutrient availability, and lake morphology. PTNs further mediated the trade-off between community diversity and productivity, providing a mechanistic link between abiotic filtering and ecosystem function. Our findings extend network theory in functional ecology to aquatic systems and offer actionable guidance for eutrophic lake restoration.

中文摘要

湖泊水生大型植物对水深和营养梯度的响应是理解湖泊功能生态学与退化湖泊修复的关键。研究在物种和群落两个尺度上分析了植物性状网络，发现局地环境条件比系统发育关系更能决定水生植物的性状配置；在群落尺度上，水深、营养水平与湖泊形态按层级顺序共同筛选群落，并影响多样性与生产力之间的权衡。

深度解读

这篇文章把“性状网络”从陆地生态扩展到了湖泊水生植物系统，是方法和应用两头都挺有价值的工作。它告诉我们，植物不是只对单一环境因子作反应，而是在一组性状协同变化中完成适应。对富营养化湖泊修复而言，这意味着不能只看某个单独指标，而应抓住与营养适应和耐受性相关的核心性状组合。换句话说，生态修复的对象不只是物种名录，更是性状结构。

DOI: https://onlinelibrary.wiley.com/doi/10.1111/gcb.70852

海洋与水生

24. The role of biofouling and microbial colonization in shaping macroplastic fate in freshwaters
（生物污损与微生物定殖如何塑造淡水中大块塑料的归趋）
Nature Water | 2026-04-08 | 【Recent】

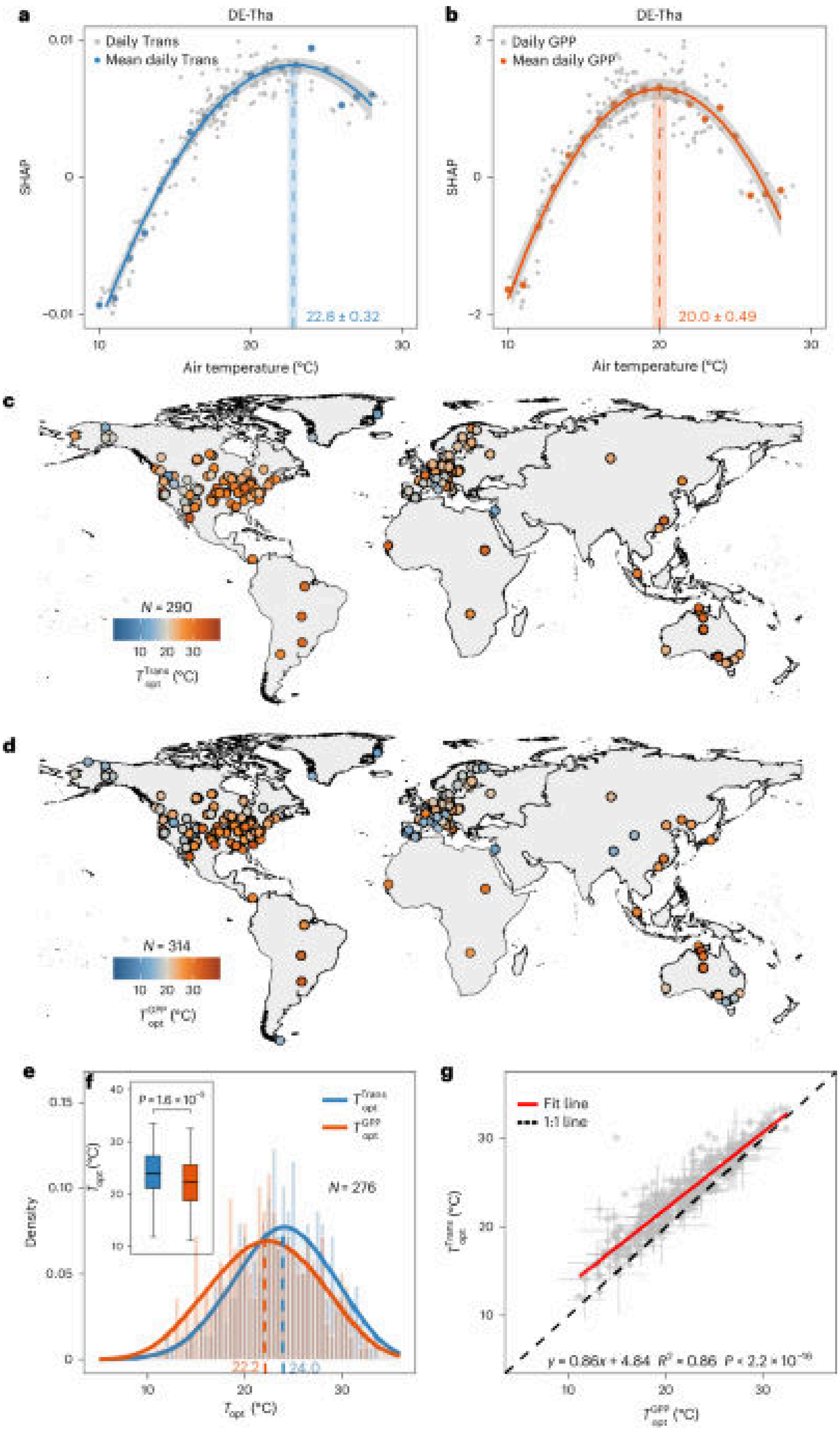


图 : Graphical abstract / article figure source: https://media.springernature.com/m685/springer-static/image/art%3A10.1038%2F

Abstract

Nature Water, Published online: 08 April 2026; doi:10.1038/s44221-026-00629-6The interaction between plastic and aquatic environments fosters a distinct microbial community on plastic surfaces known as the plastisphere. In near natural freshwater systems, microbial community composition on macroplastics is shaped primarily by water quality, while plastic substrate properties strongly influence biofilm growth and the tendency of plastics to settle.

中文摘要

塑料与水环境相互作用会形成被称为“塑料圈”的特殊微生物群落。研究在模拟城市淡水系统的 12 周中尺度实验中发现，水质决定了大块塑料上微生物群落组成，而塑料基质特性则显著影响生物膜生长和塑料下沉倾向；生物膜发育改变了材料浮力，使部分塑料更易沉降。

深度解读

这项研究很实用，因为它回答了一个管理层面经常被忽略的问题：塑料最终去哪儿了。塑料不是一直漂着不动，生物膜长上去以后，浮沉状态会发生系统变化，进而改变其生态暴露路径。文章把水质、材料性质和微生物演替放在一起看，说明塑料污染风险具有明显的环境依赖性。对河湖塑料治理而言，这意味着‘拦截漂浮物’只是第一步，沉降风险也要纳入监测。

DOI: https://doi.org/10.1038/s44221-026-00629-6

遥感与模型

25. Implementing a Dynamic Root Distribution Scheme Responsive to Soil Environment Into Noah-MP-Crop: Improving Simulations of Soil Water, Crop Growth, and Energy Fluxes in Agro-Ecosystems

（将响应土壤环境的动态根系分布方案引入 Noah-MP-Crop 以改进农田生态系统模拟）

Journal of Geophysical Research - Biogeosciences | 2026-04-08 | 【Recent】

图：Graphical abstract / article figure source:

Abstract

Reliable simulation of land – atmosphere interactions in land surface models requires realistic root distribution modeling, yet conventional static root parameterizations often fail to capture seasonal root dynamics in cropland ecosystems. This study developed a soil-environment-responsive dynamic root distribution scheme within the Noah-MP-Crop framework that explicitly accounts for soil moisture, temperature, aeration, bulk density, and soil texture. Evaluations against observations in the North China Plain showed that the new scheme outperformed default static and exponential dynamic parameterizations in simulating soil moisture, leaf area index, and latent heat flux. The simulated vertical root biomass profile also matched field observations well. Overall, the approach improves representation of root allocation, crop growth, and land – atmosphere exchanges and offers a robust pathway for coupling crop processes with climate models.

中文摘要

陆面模型中根系分布的合理表达，是准确模拟农田生态系统土壤水分、作物生长和地气交换的关键。研究在 Noah-MP-Crop 框架中构建了一个对土壤水分、温度、通气性、容重和质地敏感的动态根系分布方案，并在华北平原观测数据上验证其明显优于默认静态或指数型方案。

深度解读

这是一篇偏方法的文章，但应用意义很直接。根系分布如果写死不动，就很难真实反映作物对干旱、温度和土壤结构的响应，模型自然也会偏。新方案把根—土相互作用显式纳入后，土壤水分、叶面积指数和潜热通量模拟都更准了。对农业气候适应、灌溉优化和作物—气候耦合模拟来说，这是很实在的一步改进。

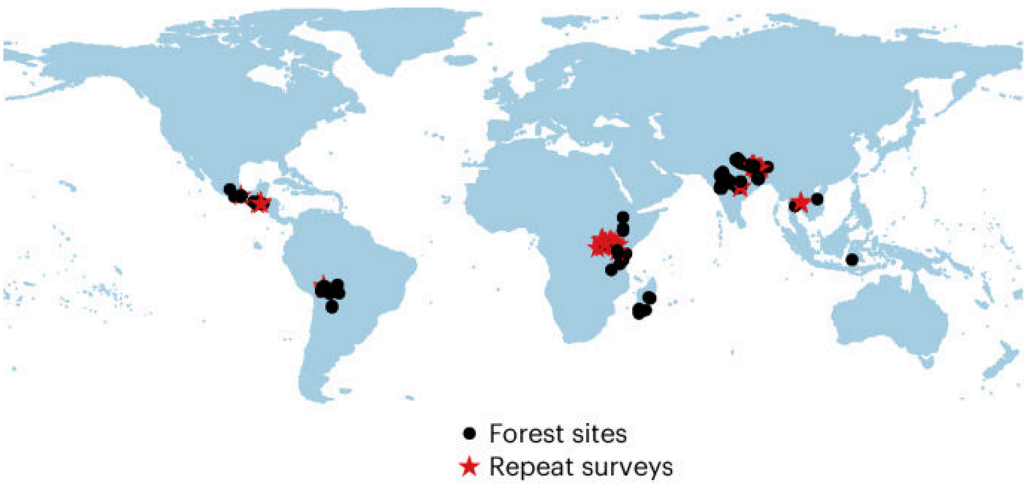
DOI: https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2025JG009462

海洋与水生

26. Multi-decadal resilience (2002–2023) of subtropical coral communities in the Solitary Islands, eastern Australia, despite severe cumulative disturbances

（尽管遭受累积性强扰动，澳大利亚东部亚热带珊瑚群落在 2002–2023 年间仍表现出长期韧性）

Coral Reefs | 2026-04-08 | 【Recent】



图：Graphical abstract / article figure source: https://static-content.springer.com/image/art%3A10.1007%2Fs00338-026-02867-2

Abstract

Environmental disturbances can cause pronounced shifts in habitat-forming species with cascading effects across marine ecosystems. Long-term monitoring is critical to detect such ecosystem change and inform management, especially in dynamic subtropical regions with frequent and cumulative, severe disturbances that are amplified by climate change. Here, we quantify multi-decadal trends (2002–2023) across coral and benthic communities at eight, shallow (9–15 m), mid-shelf (1.5–6 km offshore) rocky reefs in the Solitary Islands Marine Park (SIMP) (30 S), a biodiversity hotspot along the tropical to temperate transition zone in subtropical eastern Australia. Hard corals and algae (primarily turf and coralline algae) were the dominant taxa on these shallow mid-shelf reefs. Despite severe distur

中文摘要

基于二十余年监测数据，研究量化了澳大利亚东部 Solitary Islands Marine Park 多个浅海礁区的珊瑚和底栖群落变化。结果显示，在频繁且叠加的严重扰动背景下，这些亚热带珊瑚群落总体仍保持了显著韧性。

深度解读

在珊瑚礁研究里，长期韧性证据比单次扰动后的短期恢复更有价值。这篇文章告诉我们，并非所有边缘珊瑚系统都会一路衰退，某些亚热带群落在持续压力下仍能维持结构稳定。它为‘气候避难所’和边缘海区保护提供了希望，也提醒管理者不要只关注热带核心礁区。对未来海洋保护网络布局来说，这类具有长期韧性的过渡带海区可能值得更高优先级。

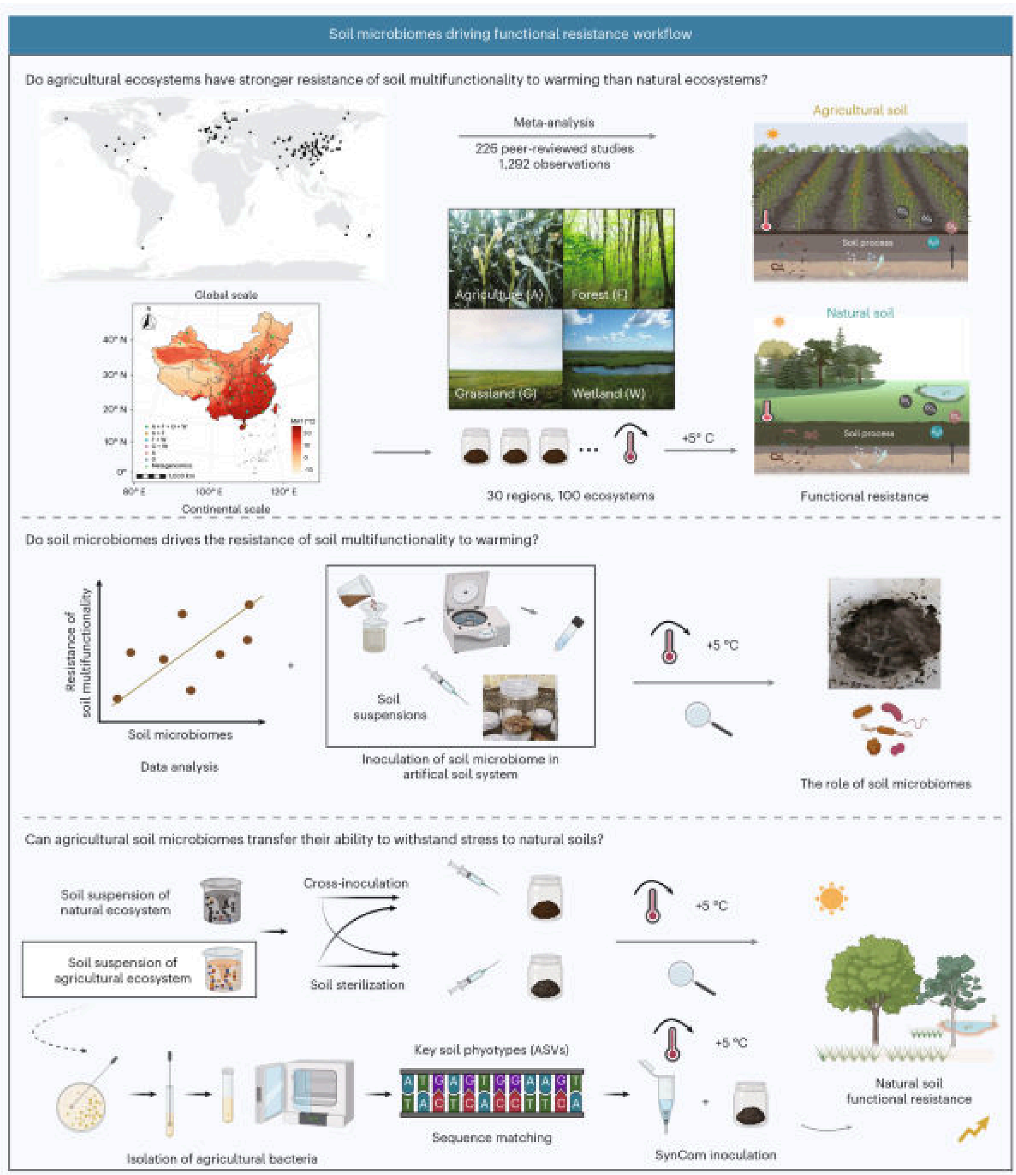
DOI: https://link.springer.com/article/10.1007/s00338-026-02867-2

保护与生物多样性

27. From species to therapeutic property: distribution differences of threatened Chinese medicinal plants under climate change

（在气候变化下，受威胁中国药用植物的分布格局及其药效类别差异正在重组）

Biodiversity and Conservation | 2026-04-16 | 【Latest】



图：Graphical abstract / article figure source: https://static-content.springer.com/image/art%3A10.1007%2Fs10531-026-03350-8

Abstract

Once medicinal plants become extinct, their unique therapeutic property will also be extinguished. At present, conservation strategies focus on overall species richness (SR), but ignore the impacts of climate change on different therapeutic categories. This study aims to explore the differences in responses of various therapeutic categories to climate change and provide scientific references for more targeted protection of medicinal resources. Using a list of 757 threatened Chinese medicinal plants (CMPs), species distribution ensemble models were used to simulate the distribution patterns of each species in the current and future (2081−2100). The analysis focused on SR and ecological niche overlap across eight therapeutic categories. It identified key environmental driving factors and del

中文摘要

研究以 757 种受威胁中国药用植物为对象，利用集成物种分布模型模拟当前与 2081−2100 年的分布变化，并比较八类治疗功效植物在物种丰富度和生态位重叠上的差异。结果表明，不同药效类别对气候变化的脆弱性并不一致，保护策略不能只看总体物种数。

深度解读

这篇文章把生物多样性保护和药用资源安全真正连到了一起。过去很多研究只问某个类群会不会减少，但这里进一步追问：如果丢失的是某类关键药效资源，社会和健康层面的代价会是什么。它说明药用植物保护不能只按物种丰富度排序，而要把功能和用途维度一起考虑。对中药资源保育、迁地保护和种质库建设，这个思路非常实际。

DOI: https://link.springer.com/article/10.1007/s10531-026-03350-8

生态学核心

28. Environmental Fluctuations Are Insufficient to Explain High Coral Biodiversity: A Test of Five Storage Effect Pathways

(环境波动不足以解释高珊瑚多样性：五种储存效应路径检验)

Ecology Letters | 2026-04-17 | 【Latest】

图：Graphical abstract / article figure source:

Abstract

中文摘要

研究围绕“环境波动不足以解释高珊瑚多样性：五种储存效应路径检验”展开。英文摘要要点为：Environmental Fluctuations Are Insufficient to Explain High Coral Biodiversity: A Test of Five Storage Effect Pathways 该条目发表于 Ecology Letters，报告中保留 DOI 与原文链接，便于后续核对全文细节。

深度解读

该研究围绕群落、种群、性状或生态系统过程，提供了可迁移的生态学机制认识。其重要性在于把具体观测或模型结果转化为可检验的机制判断。对 2026-04-18 这期速览而言，它补充了当周关于生态学核心方向的前沿证据。后续使用时应结合全文方法、空间尺度和不确定性，避免把单一区域或单一模型结论过度外推。

DOI: https://onlinelibrary.wiley.com/doi/10.1111/ele.70362

遥感与模型

29. Hydrological Mismatch in Arid Planted Shrublands: Non-Responsiveness to Precipitation Changes and Unsustainable Water Use

（干旱人工灌丛中的水文错配：对降水变化不敏感与不可持续用水）

Journal of Geophysical Research - Biogeosciences | 2026-04-17 | 【Latest】

图：Graphical abstract / article figure source:

Abstract

Journal of Geophysical Research: Biogeosciences, Volume 131, Issue 4, April 2026.

中文摘要

研究围绕“干旱人工灌丛中的水文错配：对降水变化不敏感与不可持续用水”展开。英文摘要要点为：Journal of Geophysical Research: Biogeosciences, Volume 131, Issue 4, April 2026. 该条目发表于 Journal of Geophysical Research - Biogeosciences，报告中保留 DOI 与原文链接，便于后续核对全文细节。

深度解读

该研究围绕群落、种群、性状或生态系统过程，提供了可迁移的生态学机制认识。其重要性在于把具体观测或模型结果转化为可检验的机制判断。对 2026-04-18 这期速览而言，它补充了当周关于遥感与模型方向的前沿证据。后续使用时应结合全文方法、空间尺度和不确定性，避免把单一区域或单一模型结论过度外推。

DOI: https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2026JG009715

生态学核心

30. Environmental Variability Shapes Life-History Trade-Offs Within and Between Populations of a Long-Lived Seabird

（环境变异塑造长寿海鸟种群内外的生活史权衡）

Ecology Letters | 2026-04-16 | 【Latest】

图：Graphical abstract / article figure source:

Abstract

中文摘要

研究围绕“环境变异塑造长寿海鸟种群内外的生活史权衡”展开。英文摘要要点为：Environmental Variability Shapes Life-History Trade-Offs Within and Between Populations of a Long-Lived Seabird 该条目发表于 Ecology Letters，报告中保留 DOI 与原文链接，便于后续核对全文细节。

深度解读

该研究围绕群落、种群、性状或生态系统过程，提供了可迁移的生态学机制认识。其重要性在于把具体观测或模型结果转化为可检验的机制判断。对 2026-04-18 这期速览而言，它补充了当周关于生态学核心方向的前沿证据。后续使用时应结合全文方法、空间尺度和不确定性，避免把单一区域或单一模型结论过度外推。

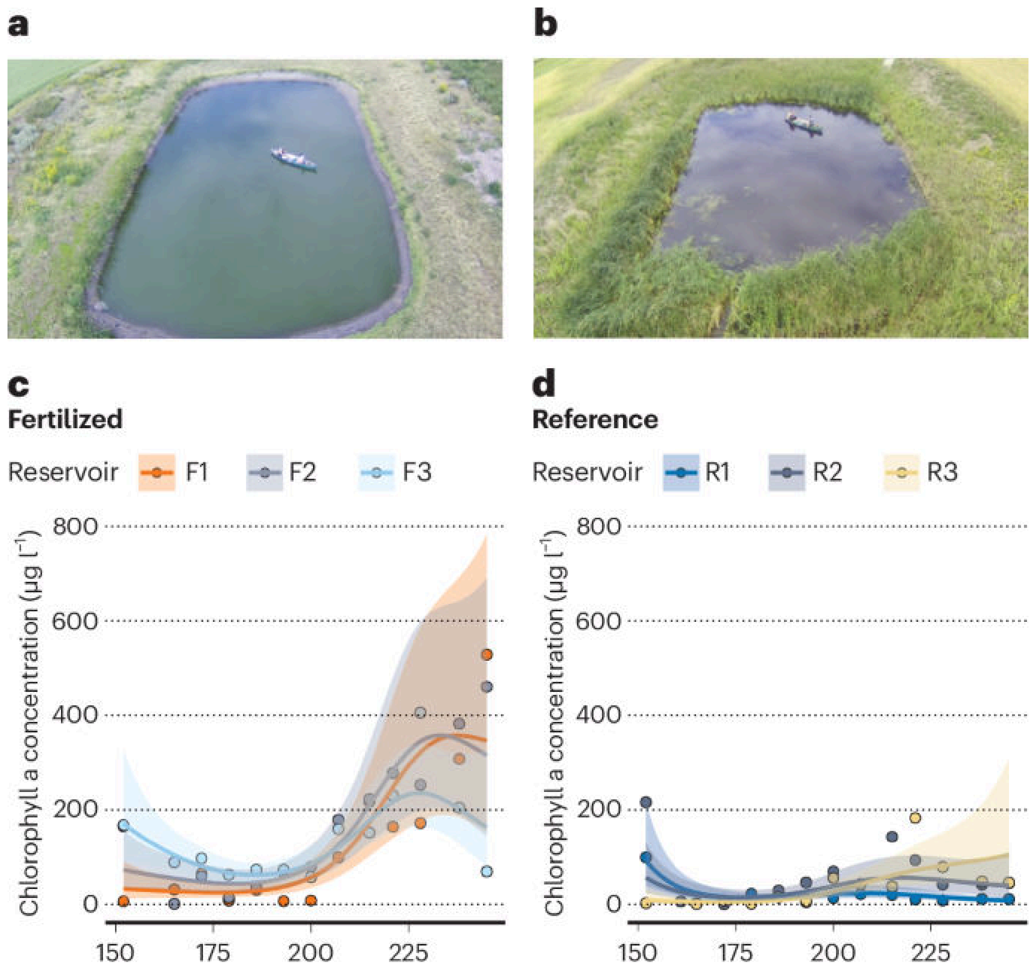
DOI: https://onlinelibrary.wiley.com/doi/10.1111/ele.70384

植物与农业

31. Losses of ecological and social sustainability among small-scale fishers in Kenya’s nearshore fisheries

（肯尼亚近岸小规模渔业中的生态与社会可持续性损失）

Nature Food | 2026-04-10 | 【Recent】



图：Graphical abstract / article figure source: https://media.springernature.com/m685/springer-static/image/art%3A10.1038%2F

Abstract

Nature Food, Published online: 10 April 2026; doi:10.1038/s43016-026-01330-3Fishers’livelihoods are threatened by the rising cost of living and dwindling fish stocks. A dataset of Kenya’s small-scale fisheries spanning 27 years reveals increasing poverty despite rising nominal fish prices and incomes, while reduced fishing effort has prompted some recovery in fish biomass.

中文摘要

研究围绕“肯尼亚近岸小规模渔业中的生态与社会可持续性损失”展开。英文摘要要点为：Fishers’ livelihoods are threatened by the rising cost of living and dwindling fish stocks. A dataset of Kenya’s small-scale fisheries spanning 27 years reveals increasing poverty despite rising nominal fish prices and incomes, while reduced fishing effort has prompted some recovery in fish biomass. 该条目发表于 Nature Food，报告中保留 DOI 与原文链接，便于后续核对全文细节。

深度解读

该研究连接植物功能、农业生产和环境胁迫，对气候适应型农业与生态系统功能评估有启发。其重要性在于把具体观测或模型结果转化为可检验的机制判断。对 2026-04-18 这期速览而言，它补充了当周关于植物与农业方向的前沿证据。后续使用时应结合全文方法、空间尺度和不确定性，避免把单一区域或单一模型结论过度外推。

DOI: https://doi.org/10.1038/s43016-026-01330-3

生态学核心

32. Advances in Modelling Radiative Transfer, Heat Storage and Turbulent Transport to Evaluate CO2, Heat and Water Fluxes Over Broad-Leaved Forests: The CanVeg2 Model

(CanVeg2 模型改进阔叶林二氧化碳、热量与水分通量模拟)

Global Change Biology | 2026-04-18 | 【Latest】

图： Graphical abstract / article figure source:

Abstract

中文摘要

研究围绕“CanVeg2 模型改进阔叶林二氧化碳、热量与水分通量模拟”展开。英文摘要要点为：Advances in Modelling Radiative Transfer, Heat Storage and Turbulent Transport to Evaluate CO2, Heat and Water Fluxes Over Broad-Leaved Forests: The CanVeg2 Model 该条目发表于 Global Change Biology，报告中保留 DOI 与原文链接，便于后续核对全文细节。

深度解读

该研究围绕群落、种群、性状或生态系统过程，提供了可迁移的生态学机制认识。其重要性在于把具体观测或模型结果转化为可检验的机制判断。对 2026-04-18 这期速览而言，它补充了当周关于生态学核心方向的前沿证据。后续使用时应结合全文方法、空间尺度和不确定性，避免把单一区域或单一模型结论过度外推。

DOI: https://onlinelibrary.wiley.com/doi/10.1111/gcb.70867

综合顶刊

33. Historical and contemporary genomes of an endangered rodent reveal shifts in environmentally associated genes

(濒危啮齿动物历史与现代基因组揭示环境关联基因变化)

Science Advances | 2026-04-17 | 【Latest】

图：Graphical abstract / article figure source:

Abstract

中文摘要

研究围绕“濒危啮齿动物历史与现代基因组揭示环境关联基因变化”展开。英文摘要要点为：Historical and contemporary genomes of an endangered rodent reveal shifts in environmentally associated genes 该条目发表于 Science Advances，报告中保留 DOI 与原文链接，便于后续核对全文细节。

深度解读

该研究围绕群落、种群、性状或生态系统过程，提供了可迁移的生态学机制认识。其重要性在于把具体观测或模型结果转化为可检验的机制判断。对 2026-04-18 这期速览而言，它补充了当周关于综合顶刊方向的前沿证据。后续使用时应结合全文方法、空间尺度和不确定性，避免把单一区域或单一模型结论过度外推。

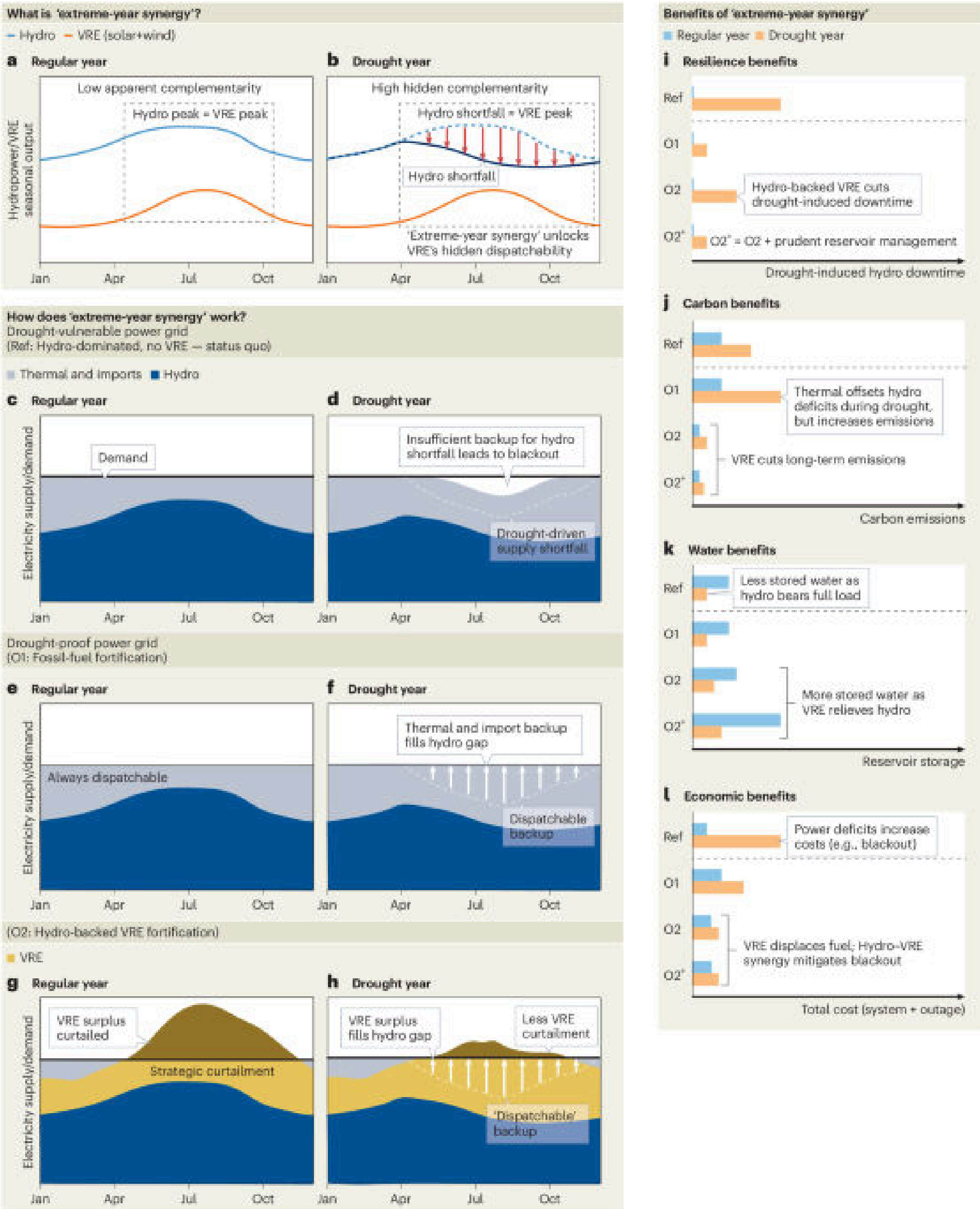
DOI: https://doi.org/10.1126/sciadv.adz5059

海洋与水生

34. A renewable tango for drought-stricken power grids

（海洋与水生研究：A renewable tango for drought-stricken power grids）

Nature Water | 2026-04-16 | 【Latest】



图：Graphical abstract / article figure source: https://media.springernature.com/m685/springer-static/image/art%3A10.1038%2F

Abstract

Nature Water, Published online: 16 April 2026; doi:10.1038/s44221-026-00625-wLike a tango partner stepping forward when the other falters, solar and wind can compensate for drought-driven hydropower shortfalls —turning these variable renewables into an underappreciated source of grid resilience.

中文摘要

研究围绕“海洋与水生研究：A renewable tango for drought-stricken power grids”展开。英文摘要要点为：Like a tango partner stepping forward when the other falters, solar and wind can compensate for drought-driven hydropower shortfalls —turning these variable renewables into an underappreciated source of grid resilience. 该条目发表于 Nature Water，报告中保留 DOI 与原文链接，便于后续核对全文细节。

深度解读

该研究把水文过程、海洋生态或淡水生态响应放在核心位置，对流域和近岸管理具有参考价值。其重要性在于把具体观测或模型结果转化为可检验的机制判断。对 2026-04-18 这期速览而言，它补充

了当周关于海洋与水生方向的前沿证据。后续使用时应结合全文方法、空间尺度和不确定性，避免把单一区域或单一模型结论过度外推。

DOI: https://doi.org/10.1038/s44221-026-00625-w

土壤与生物地球化学

35. Inorganic Carbon (Calcite and Dolomite) Determination Using the Modified Pressure-Calcimeter Method

（土壤与生物地球化学研究：Inorganic Carbon (Calcite and Dolomite) Determination Using the Modified Pressure-Calcimeter Method）

European Journal of Soil Science | 2026-04-16 | 【Latest】

图：Graphical abstract / article figure source:

Abstract

European Journal of Soil Science, Volume 77, Issue 2, March–April 2026.

中文摘要

研究围绕“土壤与生物地球化学研究：Inorganic Carbon (Calcite and Dolomite) Determination Using the Modified Pressure-Calcimeter Method”展开。英文摘要要点为：European Journal of Soil Science, Volume 77, Issue 2, March–April 2026. 该条目发表于 European Journal of Soil Science，报告中保留 DOI 与原文链接，便于后续核对全文细节。

深度解读

该研究涉及碳、氮、水分或土壤生物过程，有助于改进生态系统碳汇和养分循环判断。其重要性在于把具体观测或模型结果转化为可检验的机制判断。对 2026-04-18 这期速览而言，它补充了当周关于土壤与生物地球化学方向的前沿证据。后续使用时应结合全文方法、空间尺度和不确定性，避免把单一区域或单一模型结论过度外推。

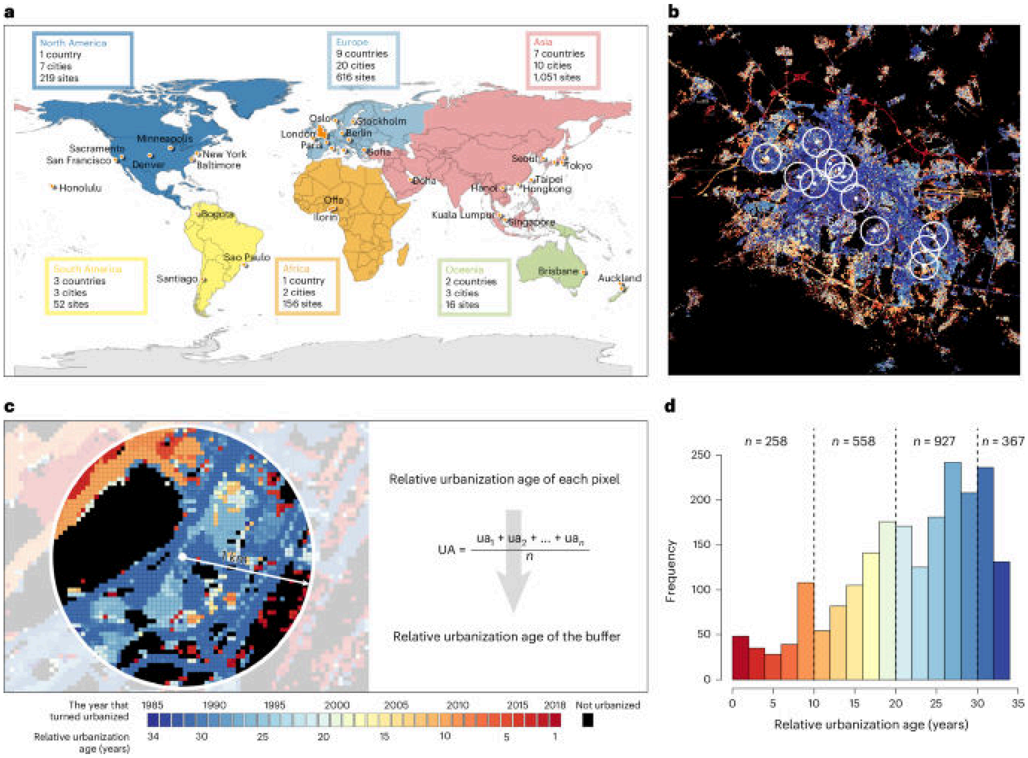
DOI: https://bsssjournals.onlinelibrary.wiley.com/doi/10.1111/ejss.70332

环境与气候

36. The linkage between microbial community dynamics and urbanization age

（微生物群落动态与城市化年限之间的联系）

Nature Sustainability | 2026-04-15 | 【Latest】



图：Graphical abstract / article figure source: https://media.springernature.com/m685/springer-static/image/art%3A10.1038%2F

Abstract

Nature Sustainability, Published online: 15 April 2026; doi:10.1038/s41893-026-01801-2Urbanization can affect the dynamics of microbial communities, which can, in turn, have implications for public health. By leveraging global metagenome datasets, the authors study the impact of urbanization age on microbial communities for actionable insights for urban planning and public health.

中文摘要

研究围绕“微生物群落动态与城市化年限之间的联系”展开。英文摘要要点为：Urbanization can affect the dynamics of microbial communities, which can, in turn, have implications for public health. By leveraging global metagenome datasets, the authors study the impact of urbanization age on microbial communities for actionable insights for urban planning and public health. 该条目发表于 Nature Sustainability，报告中保留 DOI 与原文链接，便于后续核对全文细节。

深度解读

该研究与气候变化、极端事件或环境治理直接相关，能帮助判断生态系统和人类社会面对的复合风险。其重要性在于把具体观测或模型结果转化为可检验的机制判断。对 2026-04-18 这期速览而言，它补充了当周关于环境与气候方向的前沿证据。后续使用时应结合全文方法、空间尺度和不确定性，避免把单一区域或单一模型结论过度外推。

DOI: https://doi.org/10.1038/s41893-026-01801-2

植物与农业

37. A dual sustainability crisis for Kenya’s small-scale fisheries

（肯尼亚小规模渔业面临双重可持续性危机）

Nature Food | 2026-04-14 | 【Latest】

图：Graphical abstract / article figure source: https://media.springernature.com/m685/springer-static/image/art%3A10.1038%2F

Abstract

Nature Food, Published online: 14 April 2026; doi:10.1038/s43016-026-01342-zData collected over 27 years show that ecological and social sustainability can quietly diverge, driven by factors that fisheries assessments typically overlook. For small-scale reef fishers in Kenya, rising fish prices and higher catches have failed to translate into decent earnings, with fewer than 2% making the national minimum wage.

中文摘要

研究围绕“肯尼亚小规模渔业面临双重可持续性危机”展开。英文摘要要点为：Data collected over 27 years show that ecological and social sustainability can quietly diverge, driven by factors that fisheries assessments typically overlook. For small-scale reef fishers in Kenya, rising fish prices and higher catches have failed to translate into decent earnings, with fewer than 2% making the national minimum wage. 该条目发表于 Nature Food，报告中保留 DOI 与原文链接，便于后续核对全文细节。

深度解读

该研究连接植物功能、农业生产和环境胁迫，对气候适应型农业与生态系统功能评估有启发。其重要性在于把具体观测或模型结果转化为可检验的机制判断。对 2026-04-18 这期速览而言，它补充了当周关于植物与农业方向的前沿证据。后续使用时应结合全文方法、空间尺度和不确定性，避免把单一区域或单一模型结论过度外推。

DOI: https://doi.org/10.1038/s43016-026-01342-z

生态学核心

38. Grazing Stocking Rate and Rainfall Jointly Regulate Soil Methane Uptake in a Temperate Meadow Steppe, but Do Not Reverse Net Methane Emissions

(放牧载畜率与降雨共同调节温带草甸草原土壤甲烷吸收)

Global Change Biology | 2026-04-18 | 【Latest】

图：Graphical abstract / article figure source:

Abstract

中文摘要

研究围绕“放牧载畜率与降雨共同调节温带草甸草原土壤甲烷吸收”展开。英文摘要要点为：Grazing Stocking Rate and Rainfall Jointly Regulate Soil Methane Uptake in a Temperate Meadow Steppe, but Do Not Reverse Net Methane Emissions 该条目发表于 Global Change Biology，报告中保留 DOI 与原文链接，便于后续核对全文细节。

深度解读

该研究围绕群落、种群、性状或生态系统过程，提供了可迁移的生态学机制认识。其重要性在于把具体观测或模型结果转化为可检验的机制判断。对 2026-04-18 这期速览而言，它补充了当周关于生态学核心方向的前沿证据。后续使用时应结合全文方法、空间尺度和不确定性，避免把单一区域或单一模型结论过度外推。

DOI: https://onlinelibrary.wiley.com/doi/10.1111/gcb.70864

生态学核心

39. Climate as the Main Driver of Large-Scale Genetic Patterns and Connectivity in the Expanding Golden Jackal

(气候是扩张中金豺大尺度遗传格局和连通性的主要驱动因子)

Global Change Biology | 2026-04-17 | 【Latest】

图：Graphical abstract / article figure source:

Abstract

中文摘要

研究围绕“气候是扩张中金豺大尺度遗传格局和连通性的主要驱动因子”展开。英文摘要要点为：Climate as the Main Driver of Large-Scale Genetic Patterns and Connectivity in the Expanding Golden Jackal 该条目发表于 Global Change Biology，报告中保留 DOI 与原文链接，便于后续核对全文细节。

深度解读

该研究围绕群落、种群、性状或生态系统过程，提供了可迁移的生态学机制认识。其重要性在于把具体观测或模型结果转化为可检验的机制判断。对 2026-04-18 这期速览而言，它补充了当周关于生态学核心方向的前沿证据。后续使用时应结合全文方法、空间尺度和不确定性，避免把单一区域或单一模型结论过度外推。

DOI: https://onlinelibrary.wiley.com/doi/10.1111/gcb.70879

生态学核心

40. Decoupled Climatic Drivers of Tree and Ground-Layer Carbon Uptake in Mountain Ecosystems Around the World

(全球山地生态系统乔木层与地被层碳吸收的气候驱动相互脱耦)

Global Change Biology | 2026-04-17 | 【Latest】

图：Graphical abstract / article figure source:

Abstract

中文摘要

研究围绕“全球山地生态系统乔木层与地被层碳吸收的气候驱动相互脱耦”展开。英文摘要要点为：Decoupled Climatic Drivers of Tree and Ground-Layer Carbon Uptake in Mountain Ecosystems Around the World 该条目发表于 Global Change Biology，报告中保留 DOI 与原文链接，便于后续核对全文细节。

深度解读

该研究围绕群落、种群、性状或生态系统过程，提供了可迁移的生态学机制认识。其重要性在于把具体观测或模型结果转化为可检验的机制判断。对 2026-04-18 这期速览而言，它补充了当周关于生态学核心方向的前沿证据。后续使用时应结合全文方法、空间尺度和不确定性，避免把单一区域或单一模型结论过度外推。

DOI: https://onlinelibrary.wiley.com/doi/10.1111/gcb.70877

生态学核心

41. Sexual Selection Associated With an Aggressive Male Phenotype Reduces Population Size and Hinders Population Recovery After Heat Stress

（生态学核心研究：Sexual Selection Associated With an Aggressive Male Phenotype Reduces Population Size and Hinders Population Recovery After Heat Stress）

Ecology Letters | 2026-04-17 | 【Latest】

图：Graphical abstract / article figure source:

Abstract

中文摘要

研究围绕“生态学核心研究：Sexual Selection Associated With an Aggressive Male Phenotype Reduces Population Size and Hinders Population Recovery After Heat Stress”展开。英文摘要要点为：Sexual Selection Associated With an Aggressive Male Phenotype Reduces Population Size and Hinders Population Recovery After Heat Stress 该条目发表于 Ecology Letters，报告中保留 DOI 与原文链接，便于后续核对全文细节。

深度解读

该研究围绕群落、种群、性状或生态系统过程，提供了可迁移的生态学机制认识。其重要性在于把具体观测或模型结果转化为可检验的机制判断。对 2026-04-18 这期速览而言，它补充了当周关于生态学核心方向的前沿证据。后续使用时应结合全文方法、空间尺度和不确定性，避免把单一区域或单一模型结论过度外推。

DOI: https://onlinelibrary.wiley.com/doi/10.1111/ele.70377

海洋与水生

42. Urban water network upgrades improve quality and access to drinking water in the PAASIM matched cohort study in Beira, Mozambique

（海洋与水生研究：Urban water network upgrades improve quality and access to drinking water in the PAASIM matched cohort study in Beira, Mozambique）

Nature Water | 2026-04-17 | 【Latest】

图：Graphical abstract / article figure source:

Abstract

Nature Water, Published online: 17 April 2026; doi:10.1038/s44221-026-00624-xUrban water systems in low-income cities often face contamination risks and unreliable supply, motivating evaluations of infrastructure upgrades. An assessment of the impact of water service line replacements on water quality and access in Beira, Mozambique, reveals reduced source water contamination and improved access.

中文摘要

研究围绕"海洋与水生研究：Urban water network upgrades improve quality and access to drinking water in the PAASIM matched cohort study in Beira, Mozambique"展开。英文摘要要点为：Urban water systems in low-income cities often face contamination risks and unreliable supply, motivating evaluations of infrastructure upgrades. An assessment of the impact of water service line replacements on water quality and access in Beira, Mozambique, reveals reduced source water contamination and improved access. 该条目发表于 Nature Water，报告中保留 DOI 与原文链接，便于后续核对全文细节。

深度解读

该研究把水文过程、海洋生态或淡水生态响应放在核心位置，对流域和近岸管理具有参考价值。其重要性在于把具体观测或模型结果转化为可检验的机制判断。对 2026-04-18 这期速览而言，它补充了当周关于海洋与水生方向的前沿证据。后续使用时应结合全文方法、空间尺度和不确定性，避免把单一区域或单一模型结论过度外推。

DOI: https://doi.org/10.1038/s44221-026-00624-x

遥感与模型

43. The Direct and Legacy Impacts of Droughts on Vegetation Growth in China

（干旱对中国植被生长的直接与滞后影响）

Journal of Geophysical Research - Biogeosciences | 2026-04-15 | 【Latest】

图： Graphical abstract / article figure source:

Abstract

Journal of Geophysical Research: Biogeosciences, Volume 131, Issue 4, April 2026.

中文摘要

研究围绕“干旱对中国植被生长的直接与滞后影响”展开。英文摘要要点为：Journal of Geophysical Research: Biogeosciences, Volume 131, Issue 4, April 2026. 该条目发表于 Journal of Geophysical Research - Biogeosciences，报告中保留 DOI 与原文链接，便于后续核对全文细节。

深度解读

该研究围绕群落、种群、性状或生态系统过程，提供了可迁移的生态学机制认识。其重要性在于把具体观测或模型结果转化为可检验的机制判断。对 2026-04-18 这期速览而言，它补充了当周关于遥感与模型方向的前沿证据。后续使用时应结合全文方法、空间尺度和不确定性，避免把单一区域或单一模型结论过度外推。

DOI: https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2025JG009199

植物与农业

44. Early Postfire Regeneration Determines Long-Term Successional Trajectories in Temperate Pinus densiflora Forests of Korea

（植物与农业研究：Early Postfire Regeneration Determines Long-Term Successional Trajectories in Temperate Pinus densiflora Forests of Korea）

Journal of Vegetation Science | 2026-04-13 | 【Latest】

图：Graphical abstract / article figure source:

Abstract

Journal of Vegetation Science, Volume 37, Issue 2, March/April 2026.

中文摘要

研究围绕“植物与农业研究：Early Postfire Regeneration Determines Long-Term Successional Trajectories in Temperate Pinus densiflora Forests of Korea”展开。英文摘要要点为：Journal of Vegetation Science, Volume 37, Issue 2, March/April 2026. 该条目发表于 Journal of Vegetation Science，报告中保留 DOI 与原文链接，便于后续核对全文细节。

深度解读

该研究连接植物功能、农业生产和环境胁迫，对气候适应型农业与生态系统功能评估有启发。其重要性在于把具体观测或模型结果转化为可检验的机制判断。对 2026-04-18 这期速览而言，它补充了当周关于植物与农业方向的前沿证据。后续使用时应结合全文方法、空间尺度和不确定性，避免把单一区域或单一模型结论过度外推。

DOI: https://onlinelibrary.wiley.com/doi/10.1111/jvs.70138

生态学核心

45. Multiple Mechanisms Required to Predict Grass Community Composition

（预测草地群落组成需要多个机制共同支撑）

Ecology Letters | 2026-04-12 | 【Latest】

图：Graphical abstract / article figure source:

Abstract

中文摘要

研究围绕“预测草地群落组成需要多个机制共同支撑”展开。英文摘要要点为：Multiple Mechanisms Required to Predict Grass Community Composition 该条目发表于 Ecology Letters，报告中保留 DOI 与原文链接，便于后续核对全文细节。

深度解读

该研究围绕群落、种群、性状或生态系统过程，提供了可迁移的生态学机制认识。其重要性在于把具体观测或模型结果转化为可检验的机制判断。对 2026-04-18 这期速览而言，它补充了当周关于生态学核心方向的前沿证据。后续使用时应结合全文方法、空间尺度和不确定性，避免把单一区域或单一模型结论过度外推。

DOI: https://onlinelibrary.wiley.com/doi/10.1111/ele.70358

综合顶刊

46. Performative planning creates a values mismatch between wildfire plans and community needs

（表演式规划造成野火计划与社区需求之间的价值错配）

PNAS | 2026-04-06 | 【Recent】

图：Graphical abstract / article figure source:

Abstract

SignificanceAs the size and intensity of wildfire in the United States increases, managing wildfire will require addressing multiple interconnected impacts. Most Community Wildfire Protection Plans (CWPPs) prioritize the built environment and human health,...

中文摘要

研究围绕"表演式规划造成野火计划与社区需求之间的价值错配"展开。英文摘要要点为：SignificanceAs the size and intensity of wildfire in the United States increases, managing wildfire will require addressing multiple interconnected impacts. Most Community Wildfire Protection Plans (CWPPs) prioritize the built environment and human health,... 该条目发表于 PNAS，报告中保留 DOI 与原文链接，便于后续核对全文细节。

深度解读

该研究围绕群落、种群、性状或生态系统过程，提供了可迁移的生态学机制认识。其重要性在于把具体观测或模型结果转化为可检验的机制判断。对 2026-04-18 这期速览而言，它补充了当周关于综合顶刊方向的前沿证据。后续使用时应结合全文方法、空间尺度和不确定性，避免把单一区域或单一模型结论过度外推。

DOI: https://www.pnas.org/doi/abs/10.1073/pnas.2521536123

植物与农业

47. Conserved DNA architect couples chloroplast development to cell cycle in developing cotyledons

（植物与农业研究：Conserved DNA architect couples chloroplast development to cell cycle in developing cotyledons）

Nature Plants | 2026-04-17 | 【Latest】

图：Graphical abstract / article figure source:

Abstract

Nature Plants, Published online: 17 April 2026; doi:10.1038/s41477-026-02280-1Light orchestrates cotyledon greening and growth, but the molecular link between chloroplast development and the cell cycle was unknown. Wang et al. identify RDE, a DNA-shaping protein that synchronizes these two processes in Arabidopsis.

中文摘要

研究围绕“植物与农业研究：Conserved DNA architect couples chloroplast development to cell cycle in developing cotyledons”展开。英文摘要要点为：Light orchestrates cotyledon greening and growth, but the molecular link between chloroplast development and the cell cycle was unknown. Wang et al. identify RDE, a DNA-shaping protein that synchronizes these two processes in Arabidopsis. 该条目发表于 Nature Plants，报告中保留 DOI 与原文链接，便于后续核对全文细节。

深度解读

该研究连接植物功能、农业生产和环境胁迫，对气候适应型农业与生态系统功能评估有启发。其重要性在于把具体观测或模型结果转化为可检验的机制判断。对 2026-04-18 这期速览而言，它补充了当周关于植物与农业方向的前沿证据。后续使用时应结合全文方法、空间尺度和不确定性，避免把单一区域或单一模型结论过度外推。

DOI: https://doi.org/10.1038/s41477-026-02280-1

植物与农业

48. Global food security rests on the Strait of Hormuz

（植物与农业研究：Global food security rests on the Strait of Hormuz）

Nature Food | 2026-04-17 | 【Latest】

图：Graphical abstract / article figure source:

Abstract

Nature Food, Published online: 17 April 2026; doi:10.1038/s43016-026-01340-1Global food security rests on the Strait of Hormuz

中文摘要

研究围绕“植物与农业研究：Global food security rests on the Strait of Hormuz”展开。英文摘要要点为：Global food security rests on the Strait of Hormuz 该条目发表于 Nature Food，报告中保留 DOI 与原文链接，便于后续核对全文细节。

深度解读

该研究连接植物功能、农业生产和环境胁迫，对气候适应型农业与生态系统功能评估有启发。其重要性在于把具体观测或模型结果转化为可检验的机制判断。对 2026-04-18 这期速览而言，它补充了当周关于植物与农业方向的前沿证据。后续使用时应结合全文方法、空间尺度和不确定性，避免把单一区域或单一模型结论过度外推。

DOI: https://doi.org/10.1038/s43016-026-01340-1

植物与农业

49. Stress Dominance and Niche Conservatism in Epiphytic and Lithophytic Bromeliads Along a Coastal–Inland Gradient

（植物与农业研究：Stress Dominance and Niche Conservatism in Epiphytic and Lithophytic Bromeliads Along a Coastal–Inland Gradient）

Journal of Vegetation Science | 2026-04-17 | 【Latest】

图：Graphical abstract / article figure source:

Abstract

Journal of Vegetation Science, Volume 37, Issue 2, March/April 2026.

中文摘要

研究围绕“植物与农业研究：Stress Dominance and Niche Conservatism in Epiphytic and Lithophytic Bromeliads Along a Coastal–Inland Gradient”展开。英文摘要要点为：Journal of Vegetation Science, Volume 37, Issue 2, March/April 2026. 该条目发表于 Journal of Vegetation Science，报告中保留 DOI 与原文链接，便于后续核对全文细节。

深度解读

该研究连接植物功能、农业生产和环境胁迫，对气候适应型农业与生态系统功能评估有启发。其重要性在于把具体观测或模型结果转化为可检验的机制判断。对 2026-04-18 这期速览而言，它补充了当周关于植物与农业方向的前沿证据。后续使用时应结合全文方法、空间尺度和不确定性，避免把单一区域或单一模型结论过度外推。

DOI: https://onlinelibrary.wiley.com/doi/10.1111/jvs.70137

生态学核心

50. Group Size Dependent Selection for Cooperation Versus Freeloading in Collective Chemical Defence

（生态学核心研究：Group Size Dependent Selection for Cooperation Versus Freeloading in Collective Chemical Defence）

Ecology Letters | 2026-04-16 | 【Latest】

图： Graphical abstract / article figure source:

Abstract

中文摘要

研究围绕“生态学核心研究：Group Size Dependent Selection for Cooperation Versus Freeloading in Collective Chemical Defence”展开。英文摘要要点为：Group Size Dependent Selection for Cooperation Versus Freeloading in Collective Chemical Defence 该条目发表于 Ecology Letters，报告中保留 DOI 与原文链接，便于后续核对全文细节。

深度解读

该研究围绕群落、种群、性状或生态系统过程，提供了可迁移的生态学机制认识。其重要性在于把具体观测或模型结果转化为可检验的机制判断。对 2026-04-18 这期速览而言，它补充了当周关于生态学核心方向的前沿证据。后续使用时应结合全文方法、空间尺度和不确定性，避免把单一区域或单一模型结论过度外推。

DOI: https://onlinelibrary.wiley.com/doi/10.1111/ele.70378

植物与农业

51. Greening the scars

（植物与农业研究：Greening the scars）

Nature Plants | 2026-04-15 | 【Latest】

图： Graphical abstract / article figure source:

Abstract

Nature Plants, Published online: 15 April 2026; doi:10.1038/s41477-026-02287-8Greening the scars

中文摘要

研究围绕“植物与农业研究：Greening the scars”展开。英文摘要要点为：Greening the scars 该条目发表于 Nature Plants，报告中保留 DOI 与原文链接，便于后续核对全文细节。

深度解读

该研究连接植物功能、农业生产和环境胁迫，对气候适应型农业与生态系统功能评估有启发。其重要性在于把具体观测或模型结果转化为可检验的机制判断。对 2026-04-18 这期速览而言，它补充了当周关于植物与农业方向的前沿证据。后续使用时应结合全文方法、空间尺度和不确定性，避免把单一区域或单一模型结论过度外推。

DOI: https://doi.org/10.1038/s41477-026-02287-8

植物与农业

52. Climatically Driven Stability in the Alpine/Moorland Boundary on Three Mountains in Southwest Lutruwita/Tasmania 1986–2020

（植物与农业研究：Climatically Driven Stability in the Alpine/Moorland Boundary on Three Mountains in Southwest Lutruwita/Tasmania 1986–2020）

Journal of Vegetation Science | 2026-04-14 | 【Latest】

图：Graphical abstract / article figure source:

Abstract

Journal of Vegetation Science, Volume 37, Issue 2, March/April 2026.

中文摘要

研究围绕"植物与农业研究：Climatically Driven Stability in the Alpine/Moorland Boundary on Three Mountains in Southwest Lutruwita/Tasmania 1986–2020"展开。英文摘要要点为：Journal of Vegetation Science, Volume 37, Issue 2, March/April 2026. 该条目发表于 Journal of Vegetation Science，报告中保留 DOI 与原文链接，便于后续核对全文细节。

深度解读

该研究连接植物功能、农业生产和环境胁迫，对气候适应型农业与生态系统功能评估有启发。其重要性在于把具体观测或模型结果转化为可检验的机制判断。对 2026-04-18 这期速览而言，它补充了当周关于植物与农业方向的前沿证据。后续使用时应结合全文方法、空间尺度和不确定性，避免把单一区域或单一模型结论过度外推。

DOI: https://onlinelibrary.wiley.com/doi/10.1111/jvs.70136

保护与生物多样性

53. Benthic diatoms and phytoplankton diversity in urban streams and ponds: charting a course for conservation

（城市溪流与池塘中底栖硅藻和浮游植物多样性的保护启示）

Biodiversity and Conservation | 2026-04-17 | 【Latest】

图：Graphical abstract / article figure source:

Abstract

Benthic diatom and phytoplankton communities are key indicators of freshwater ecosystem health and change but remain understudied. A better understanding of their distribution and abundance drivers is crucial for effective biodiversity assessment and conservation. This study adopts a novel, multi-ecosystem approach to examine the seasonal (2019–2020) diversity of benthic diatom assemblages in urban streams and phytoplankton assemblages in ponds across Bristol (UK) and the local environmental factors shaping these communities. Among ponds, winter recorded significantly fewer phytoplankton taxa than summer and autumn. Among streams, patterns of benthic diatom beta diversity indicated higher turnover in autumn than in winter, and among pond habitats, turnover dominated in the summer and autum

中文摘要

研究围绕"城市溪流与池塘中底栖硅藻和浮游植物多样性的保护启示"展开。英文摘要要点为：Benthic diatom and phytoplankton communities are key indicators of freshwater ecosystem health and change but remain understudied. A better understanding of their distribution and abundance drivers is crucial for effective biodiversity assessment and conservation. This study adopts a novel, multi-ecosystem approach to examine the seasonal (2019–2020) diversity of benthic diatom assemblages in urban streams and phytoplankton assemblages in ponds across Bristol (UK) and the local environmental factors shaping these communities. Among ponds, winter recorded significantly fewer phytoplankton taxa than summer and autumn. Among streams, patterns of benthic diatom beta diversity indicated higher turnover in autumn than in winter, and among pond habitats, turnover dominated in the summer and autum 该条目发表于 Biodiversity and Conservation，报告中保留 DOI 与原文链接，便于后续核对全文细节。

深度解读

该研究关注物种、栖息地或保护成效，强调生物多样性风险需要与土地利用和气候变化共同评估。其重要性在于把具体观测或模型结果转化为可检验的机制判断。对 2026-04-18 这期速览而言，它补充了当周关于保护与生物多样性方向的前沿证据。后续使用时应结合全文方法、空间尺度和不确定性，避免把单一区域或单一模型结论过度外推。

DOI: https://link.springer.com/article/10.1007/s10531-026-03305-z

----

海洋与水生

54. Thermal stress restructures volatile gas emissions from the model cnidarian Aiptasia

（热胁迫重构模式刺胞动物 Aiptasia 的挥发性气体排放）

Coral Reefs | 2026-04-08 | 【Recent】

图： Graphical abstract / article figure source:

Abstract

Coral bleaching events, in which symbionts are lost from host tissues, have become more frequent and severe because of climate change—specifically, elevated temperatures. How such events impact biogenic volatile organic compound (BVOC) emissions, compounds that can function as metabolic signalling elements, remains underexplored. Here we used gas chromatography coupled with mass spectrometry (GC−MS) to characterise the suite of BVOCs (collectively the 'volatilome') from the model sea anemone Exaiptasia diaphana ('Aiptasia') under three temperatures (control: 25 °C; sub-bleaching: 30 °C; and bleaching: 33.5 °C), both without symbionts (aposymbiotic) and when populated by its native dinoflagellate symbiont, Breviolum minutum. The volatilome of symbiotic anemones during bleaching at an elevat

中文摘要

研究围绕“热胁迫重构模式刺胞动物 Aiptasia 的挥发性气体排放”展开。英文摘要要点为：Coral bleaching events, in which symbionts are lost from host tissues, have become more frequent and severe because of climate change—specifically, elevated temperatures. How such events impact biogenic volatile organic compound (BVOC) emissions, compounds that can function as metabolic signalling elements, remains underexplored. Here we used gas chromatography coupled with mass spectrometry (GC−MS) to characterise the suite of BVOCs (collectively the ‘volatilome’) from the model sea anemone Exaiptasia diaphana (‘Aiptasia’) under three temperatures (control: 25 °C; sub-bleaching: 30 °C; and bleaching: 33.5 °C), both without symbionts (aposymbiotic) and when populated by its native dinoflagellate symbiont, Breviolum minutum. The volatilome of symbiotic anemones during bleaching at an elevat 该条目发表于 Coral Reefs，报告中保留 DOI 与原文链接，便于后续核对全文细节。

深度解读

该研究把水文过程、海洋生态或淡水生态响应放在核心位置，对流域和近岸管理具有参考价值。其重要性在于把具体观测或模型结果转化为可检验的机制判断。对 2026-04-18 这期速览而言，它补充了当周关于海洋与水生方向的前沿证据。后续使用时应结合全文方法、空间尺度和不确定性，避免把单一区域或单一模型结论过度外推。

DOI: https://link.springer.com/article/10.1007/s00338-026-02866-3

海洋与水生

55. Drone imaging can accurately assess coral cover, bleaching, and growth form for shallow coral reefs

(无人机影像可准确评估浅水珊瑚礁覆盖度、白化和生长型)

Coral Reefs | 2026-04-09 | 【Recent】

图： Graphical abstract / article figure source:

Abstract

Assessing the impacts of rapid environmental change on coral reefs is hindered by a discrepancy between the regions with the greatest need and those that receive the most research funding. Remote sensing, particularly through the use of drones [or unoccupied aerial vehicles (UAVs)], has the potential to significantly enhance the accessibility and efficiency of high-quality data collection in remote, biodiverse areas such as the Coral Triangle. To test this, we compared a UAV-based method, a common citizen science method (Reef Check), and a research-grade method (photo quadrats) to assess hard coral cover, coral bleaching, and coral growth forms across reef types (nearshore, fringing, midshelf) and depths (crest, 5 m, 10 m, 15 m) in Kimbe Bay, Papua New Guinea, during the fourth Global Mass

中文摘要

研究围绕“无人机影像可准确评估浅水珊瑚礁覆盖度、白化和生长型”展开。英文摘要要点为: Assessing the impacts of rapid environmental change on coral reefs is hindered by a discrepancy between the regions with the greatest need and those that receive the most research funding. Remote sensing, particularly through the use of drones [or unoccupied aerial vehicles (UAVs)], has the potential to significantly enhance the accessibility and efficiency of high-quality data collection in remote, biodiverse areas such as the Coral Triangle. To test this, we compared a UAV-based method, a common citizen science method (Reef Check), and a research-grade method (photo quadrats) to assess hard coral cover, coral bleaching, and coral growth forms across reef

types (nearshore, fringing, midshelf) and depths (crest, 5 m, 10 m, 15 m) in Kimbe Bay, Papua New Guinea, during the fourth Global Mass 该条目发表于 Coral Reefs，报告中保留 DOI 与原文链接，便于后续核对全文细节。

深度解读

该研究把水文过程、海洋生态或淡水生态响应放在核心位置，对流域和近岸管理具有参考价值。其重要性在于把具体观测或模型结果转化为可检验的机制判断。对 2026-04-18 这期速览而言，它补充了当周关于海洋与水生方向的前沿证据。后续使用时应结合全文方法、空间尺度和不确定性，避免把单一区域或单一模型结论过度外推。

DOI: https://link.springer.com/article/10.1007/s00338-026-02864-5

环境与气候

56. Increasing extremity and accelerating expansion of heatwaves across Eurasia

（欧亚大陆热浪极端性增强且扩张加速）

Environmental Research Letters | 2026-04-16 | 【Latest】

图：Graphical abstract / article figure source:

Abstract

Heatwaves have increased globally under climate warming. However, quantitative assessments regarding their spatial expansion and extremity evolution at the continental scale remain limited. In this study, we used multiple datasets and a unified framework to assess long-term changes in heatwave frequency, intensity and spatial clustering across 15 reference regions of the Intergovernmental Panel on Climate Change in Eurasia from 1950 to 2023. In Eurasia, heatwaves have shifted toward more extreme conditions since the 1990s. The spatial extent of rare events, which represent the most extreme conditions exceeding the 90th percentile of the 1991–2020 baseline period, expanded most rapidly. The annual mean area affected by rare heatwaves during 2014–2023 has tripled relative to the climatologic

中文摘要

研究围绕“欧亚大陆热浪极端性增强且扩张加速”展开。英文摘要要点为：Heatwaves have increased globally under climate warming. However, quantitative assessments regarding their spatial expansion and extremity evolution at the continental scale remain limited. In this study, we used multiple datasets and a unified framework to assess long-term changes in heatwave frequency, intensity and spatial clustering across 15 reference regions of the Intergovernmental Panel on Climate Change in Eurasia from 1950 to 2023. In Eurasia, heatwaves have shifted toward more extreme conditions since the 1990s. The spatial extent of rare events, which represent the most extreme conditions exceeding the 90th percentile of the 1991–2020 baseline period, expanded most rapidly. The annual mean area affected by rare heatwaves during 2014–2023 has tripled relative to the climatologic 该条目发表于 Environmental Research Letters，报告中保留 DOI 与原文链接，便于后续核对全文细节。

深度解读

该研究与气候变化、极端事件或环境治理直接相关，能帮助判断生态系统和人类社会面对的复合风险。其重要性在于把具体观测或模型结果转化为可检验的机制判断。对 2026-04-18 这期速览而言，它补充了当周关于环境与气候方向的前沿证据。后续使用时应结合全文方法、空间尺度和不确定性，避免把单一区域或单一模型结论过度外推。

DOI: https://iopscience.iop.org/article/10.1088/1748-9326/ae5990

保护与生物多样性

57. Unveiling shifting frontiers of tropical dry forests depends on consistently accurate delineation of their distribution: remarks about Rivas et al. (2025) and others

（保护与生物多样性研究：Unveiling shifting frontiers of tropical dry forests depends on consistently accurate delineation of their distribution: remarks about Rivas et al. (2025) and others）

Biodiversity and Conservation | 2026-04-17 | 【Latest】

图：Graphical abstract / article figure source:

Abstract

Recently in Biodiversity and Conservation, it was asserted that 9.64% and 5.96% of tropical dry forests (TDFs) in South America and Africa were lost between 2000 and 2020. Arriving at such results depends on how TDFs are defined and how their global distribution is mapped. A striking feature of recent maps of TDFs is the substantial overlap between them and maps of tropical grassy ecosystems (TGEs). Alarmingly, both types of maps are generally based on just two earlier maps of global terrestrial ecosystems. If extensive areas of global TGEs are being misclassified as TDFs, then there are problems because policies relevant to forests, e.g., fire suppression and ‘reforestation’are not applicable to TGEs. Mapping of TGEs typically relies on ecologically functional criteria such as the distri

中文摘要

研究围绕“保护与生物多样性研究：Unveiling shifting frontiers of tropical dry forests depends on consistently accurate delineation of their distribution: remarks about Rivas et al. (2025) and others”展开。英文摘要要点为：Recently in Biodiversity and Conservation, it was asserted that 9.64% and 5.96% of tropical dry forests (TDFs) in South America and Africa were lost between 2000 and 2020. Arriving at such results depends on how TDFs are defined and how their global distribution is mapped. A striking feature of recent maps of TDFs is the substantial overlap between them and maps of tropical grassy ecosystems (TGEs). Alarmingly, both types of maps are generally based on just two earlier maps of global terrestrial ecosystems. If extensive areas of global TGEs are being misclassified as TDFs, then there are problems because policies relevant to forests, e.g., fire suppression and ‘reforestation’are not applicable to TGEs. Mapping of TGEs typically relies on ecologically functional criteria such as the distri 该条目发表于 Biodiversity and Conservation，报告中保留 DOI 与原文链接，便于后续核对全文细节。

深度解读

该研究关注物种、栖息地或保护成效，强调生物多样性风险需要与土地利用和气候变化共同评估。其重要性在于把具体观测或模型结果转化为可检验的机制判断。对 2026-04-18 这期速览而言，它补充了当周关于保护与生物多样性方向的前沿证据。后续使用时应结合全文方法、空间尺度和不确定性，避免把单一区域或单一模型结论过度外推。

DOI: https://link.springer.com/article/10.1007/s10531-026-03343-7

保护与生物多样性

58. Robots for ecological monitoring: a review

（保护与生物多样性研究：Robots for ecological monitoring: a review）

Biodiversity and Conservation | 2026-04-16 | 【Latest】

图：Graphical abstract / article figure source:

Abstract

Technologies for biodiversity assessment are crucial for addressing the global biodiversity crisis, and ecological monitoring is seeing increased use of robotics. Interdisciplinary working will be critical in finding the best robotic solutions for monitoring. As a team of ecologists and robotics scientists, we conducted a systematic literature search for robots in ecological monitoring and reviewed their ecological monitoring applications and technological capabilities, such as level of autonomy, operation time and adaptability. We identified 67 relevant papers reporting individual robots or robot teams. The number of robots developed for ecological monitoring has grown rapidly in recent years, targeting a wide range of taxa or communities. Image or video capture were the most common types

中文摘要

研究围绕“保护与生物多样性研究：Robots for ecological monitoring: a review”展开。英文摘要要点为：Technologies for biodiversity assessment are crucial for addressing the global biodiversity crisis, and ecological monitoring is seeing increased use of robotics. Interdisciplinary working will be critical in finding the best robotic solutions for monitoring. As a team of ecologists and robotics scientists, we conducted a systematic literature search for robots in ecological monitoring and reviewed their ecological monitoring applications and technological capabilities, such as level of autonomy, operation time and adaptability. We identified 67 relevant papers reporting individual robots or robot teams. The number of robots developed for ecological monitoring has grown rapidly in recent years, targeting a wide range of taxa or communities. Image or video capture were the most common types 该条目发表于 Biodiversity and Conservation，报告中保留 DOI 与原文链接，便于后续核对全文细节。

深度解读

该研究关注物种、栖息地或保护成效，强调生物多样性风险需要与土地利用和气候变化共同评估。其重要性在于把具体观测或模型结果转化为可检验的机制判断。对 2026-04-18 这期速览而言，它补充了当周关于保护与生物多样性方向的前沿证据。后续使用时应结合全文方法、空间尺度和不确定性，避免把单一区域或单一模型结论过度外推。

DOI: https://link.springer.com/article/10.1007/s10531-026-03339-3

海洋与水生

59. Fine-scale aquatic eDNA sampling reveals significant within- and across-site variation during the expected breeding season of Cryptobranchus alleganiensis alleganiensis

（海洋与水生研究：Fine-scale aquatic eDNA sampling reveals significant within- and across-site variation during the expected breeding season of Cryptobranchus alleganiensis alleganiensis）

Hydrobiologia | 2026-04-15 | 【Latest】

图： Graphical abstract / article figure source:

Abstract

Environmental DNA (eDNA) analysis is a powerful tool for monitoring wildlife populations. While seasonal peaks in eDNA concentration during breeding seasons of target aquatic species are well documented, fine-scale temporal changes in eDNA are yet to be extensively examined. We used C. a. alleganiensis, an aquatic salamander whose breeding season is characterized by high eDNA concentrations, as a model species to test three hypotheses: eDNA concentration would remain high throughout the expected breeding season, eDNA signatures would reflect synchronous breeding among local populations, and eDNA concentration changes would correspond to seasonal stream-temperature changes. We sampled five tributaries of the Susquehanna River in Pennsylvania, USA, daily from August to October to examine how

中文摘要

研究围绕“海洋与水生研究：Fine-scale aquatic eDNA sampling reveals significant within- and across-site variation during the expected breeding season of Cryptobranchus alleganiensis alleganiensis”展开。英文摘要要点为：Environmental DNA (eDNA) analysis is a powerful tool for monitoring wildlife populations. While seasonal peaks in eDNA concentration during breeding seasons of target aquatic species are well documented, fine-scale temporal changes in eDNA are yet to be extensively examined. We used C. a. alleganiensis, an aquatic salamander whose breeding season is characterized by high eDNA concentrations, as a model species to test three hypotheses: eDNA concentration would remain high throughout the expected breeding season, eDNA signatures would reflect synchronous breeding among local populations, and eDNA concentration changes would correspond to seasonal stream-temperature changes. We sampled five tributaries of the Susquehanna River in Pennsylvania, USA, daily from August to October to examine how 该条目发表于 Hydrobiologia，报告中保留 DOI 与原文链接，便于后续核对全文细节。

深度解读

该研究把水文过程、海洋生态或淡水生态响应放在核心位置，对流域和近岸管理具有参考价值。其重要性在于把具体观测或模型结果转化为可检验的机制判断。对 2026-04-18 这期速览而言，它补充了当周关于海洋与水生方向的前沿证据。后续使用时应结合全文方法、空间尺度和不确定性，避免把单一区域或单一模型结论过度外推。

DOI: https://link.springer.com/article/10.1007/s10750-026-06186-z

----

海洋与水生

60. Sixty years of artificial lake heating impact Daphnia fitness but not parasite resistance

（海洋与水生研究：Sixty years of artificial lake heating impact Daphnia fitness but not parasite resistance）

Hydrobiologia | 2026-04-15 | 【Latest】

图：Graphical abstract / article figure source:

Abstract

The impacts of climate change are expected to be complex and severe, with rising temperatures potentially exacerbating the prevalence of infectious diseases. Aquatic ectotherms, such as Daphnia, are particularly vulnerable to temperature fluctuations, which can significantly impact their fitness and disease resistance. Using a unique system of lakes, some of which have been artificially heated for over 60 years, we investigated how elevated temperatures have affected fitness and parasite resistance in Daphnia longispina. Individuals from heated lakes as well as from nearby control lakes were experimentally exposed to the allopatric fungal parasite Metschnikowia biscuspidata at two temperatures (18 °C and 22 °C). We found that Daphnia exhibited high resistance to Metschnikowia, regardless o

中文摘要

研究围绕“海洋与水生研究：Sixty years of artificial lake heating impact Daphnia fitness but not parasite resistance”展开。英文摘要要点为：The impacts of climate change are expected to be complex and severe, with rising temperatures potentially exacerbating the prevalence of infectious diseases. Aquatic ectotherms, such as Daphnia, are particularly vulnerable to temperature fluctuations, which can significantly impact their fitness and disease resistance. Using a unique system of lakes, some of which have been artificially heated for over 60 years, we investigated how elevated temperatures have affected fitness and parasite resistance in Daphnia longispina. Individuals from heated lakes as well as from nearby control lakes were experimentally exposed to the allopatric fungal parasite Metschnikowia biscuspidata at two temperatures (18 °C and 22 °C). We found that Daphnia exhibited high resistance to Metschnikowia, regardless o 该条目发表于 Hydrobiologia，报告中保留 DOI 与原文链接，便于后续核对全文细节。

深度解读

该研究把水文过程、海洋生态或淡水生态响应放在核心位置，对流域和近岸管理具有参考价值。其重要性在于把具体观测或模型结果转化为可检验的机制判断。对 2026-04-18 这期速览而言，它补充了当周关于海洋与水生方向的前沿证据。后续使用时应结合全文方法、空间尺度和不确定性，避免把单一区域或单一模型结论过度外推。

DOI: https://link.springer.com/article/10.1007/s10750-025-06106-7
